Supporting Information for

Making the Most of Little Time: A Probabilistic Approach to Marine Heatwave Detection

Sina S. Pinter[1, 2], Matthew D. Rayson[1, 2], and Nicole L. Jones[1, 2]

1 School of Earth and Oceans, The University of Western Australia, Western Australia, Australia
2 Oceans Institute, The University of Western Australia, Western Australia, Australia

Contents of this file

Figure M.1.:	Workflow of the Bayesian Parameter Inference Framework for MHW Detection
Methods:	Extension of the Bayesian Framework to Include a Trend Component
Figure M.2. and Methods:	Sensitivity of HPS-based MHW Event Detection Parameters.
Table T1:	MCMC Diagnostics Summary
Figures S0.1. to 0.3.:	Additional Figure for Site South WA
Figure S0.4.:	Example of probabilistic detection applied to a neutral five year window without major extreme events for Site South WA
Figures S1.1. to 1.9.:	Figures for Site Japan
Figures S2.1. to 2.9.:	Figures for Site North West Shelf, Australia
Figures S3.1. to 3.9.:	Figures for Site UK
Figures S4.1. to 4.9.:	Figures for Site South WA 50m

Introduction

This Supporting Information provides a workflow chart containing step by step instructions for users on how to implement the framework, as well as details on an optional extension of the Bayesian framework to include a trend component, and a sensitivity analysis evaluating the influence of HPS based MHW detection parameters on the identified event characteristics. A summary table of MCMC diagnostics is also presented to confirm model convergence. Furthermore, this document includes an evaluation of the framework applied to a neutral five year period without major extreme events to illustrate baseline variance behaviour. Finally, additional diagnostic and model output figures are provided for the South WA, Japan, North West Shelf, UK, and South WA 50m (mooring WATR10) sites. These figures are equivalent to those presented in the main text for the South WA SST site and are included here to demonstrate the consistency of model behaviour and sampling diagnostics across all study locations.

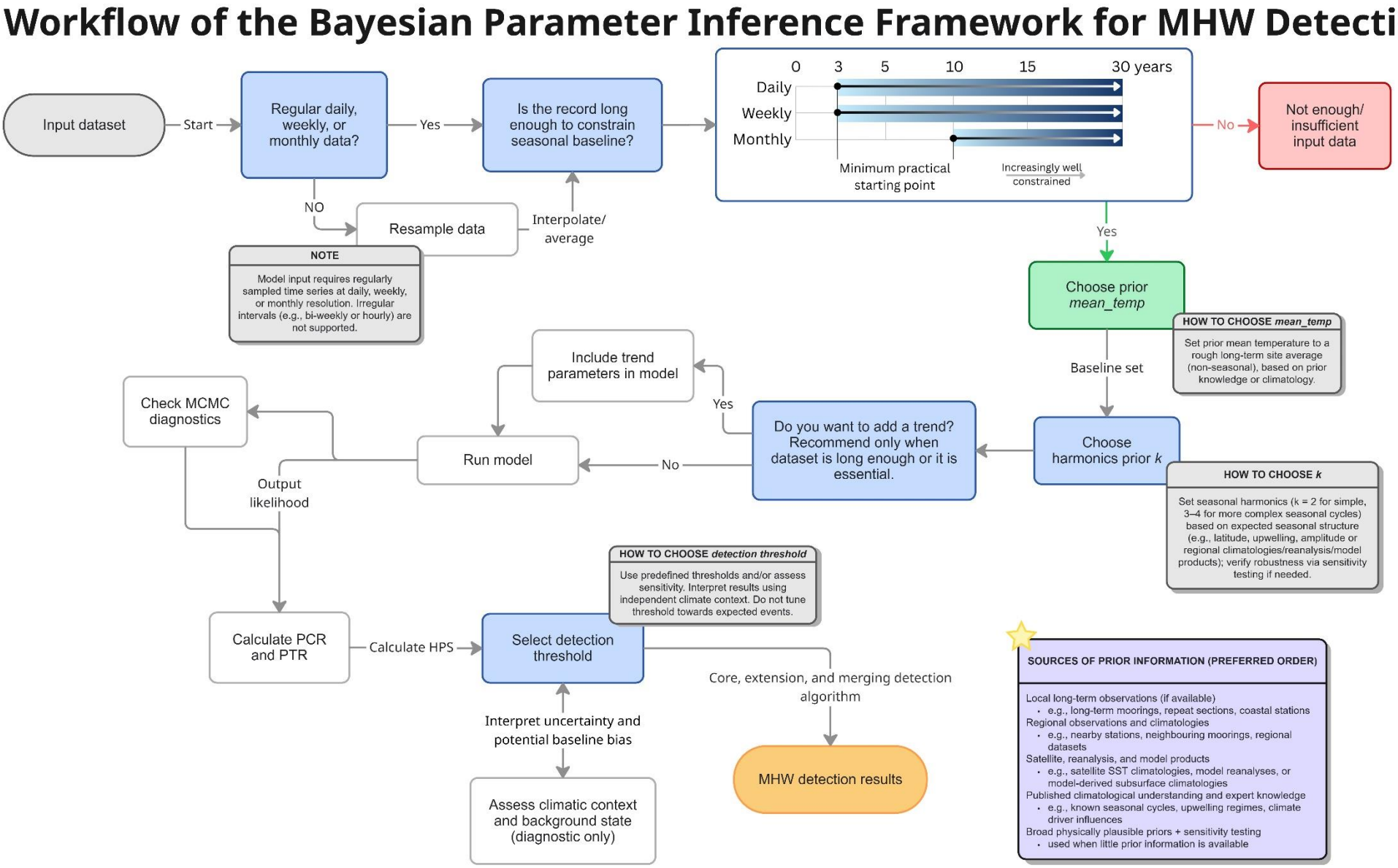


Figure M.1. Workflow of the Bayesian Parameter Inference Framework for MHW Detection

Methods: Extension of the Bayesian Framework to Include a Trend Component

Although the main analysis excludes a trend component due to the limited length of the observational records, the framework can be extended to include a linear trend term when longer records are available or when prior knowledge indicates that a trend is meaningful. In this case, the temperature decomposition becomes:

$$T(t) = T_{trend}(t) + T_s(t) + T_{non-s}$$

where the trend component is represented as:

$$T_{trend}(t) = T_{mean} + m * t$$

Wheras T_{mean} is as described in the main manuscript and m represents the slope of the trend. The trend parameter is estimated within the Bayesian framework using a normal prior:

$$m = \mathcal{N}(\mu_m, \sigma_m)$$

where σ_m and μ_m define the prior expectation and uncertainty of the trend magnitude, respectively. These parameters can be adjusted depending on the available record length and prior knowledge of the system, with tighter constraints appropriate when independent information on the expected trend exists.

Including a trend component allows the probabilistic climatology and threshold estimates (PCR and PTR) to account for gradual long-term changes in the temperature baseline, while still propagating uncertainty associated with the estimated trend.

Methods: Sensitivity of HPS-based MHW Event Detection Parameters

To evaluate the robustness of the HPS-based MHW detection algorithm, we tested the sensitivity of detected event characteristics to variations in the main detection parameters. The default configuration used in the manuscript (detection threshold=50%, ramp threshold=20%, max_gap=5 days, and merge_gap=5 days) was compared against a range of alternative parameter combinations. All using a min_duration = 5 days.

The influence of the HPS detection threshold was evaluated by varying the threshold from 10% to 90% in 10% increments, while all other parameters were held fixed at their default values. Similarly, the influence of the ramp threshold was assessed by varying the ramp threshold between 5% and 50% in 5% increments, while the detection threshold was fixed at its default value of 50% and all continuity parameters were held constant. To evaluate the sensitivity to event continuity assumptions, max_gap and merge_gap were independently varied between 1 and 20 days, with all other parameters fixed at their default settings.

For each parameter combination, the HPS-based detection algorithm was applied to the same South WA HPS time series shown in Figure 3 of the main manuscript. The resulting detected events were compared based on the number of identified MHW events and total MHW duration. In addition, detected events were compared against the default configuration by calculating the temporal overlap of identified MHW periods to assess whether alternative parameter choices resulted in substantially different event timing.

This sensitivity analysis evaluates whether the detected MHW characteristics depend strongly on individual parameter choices and provides an assessment of the robustness of the probabilistic event detection framework.

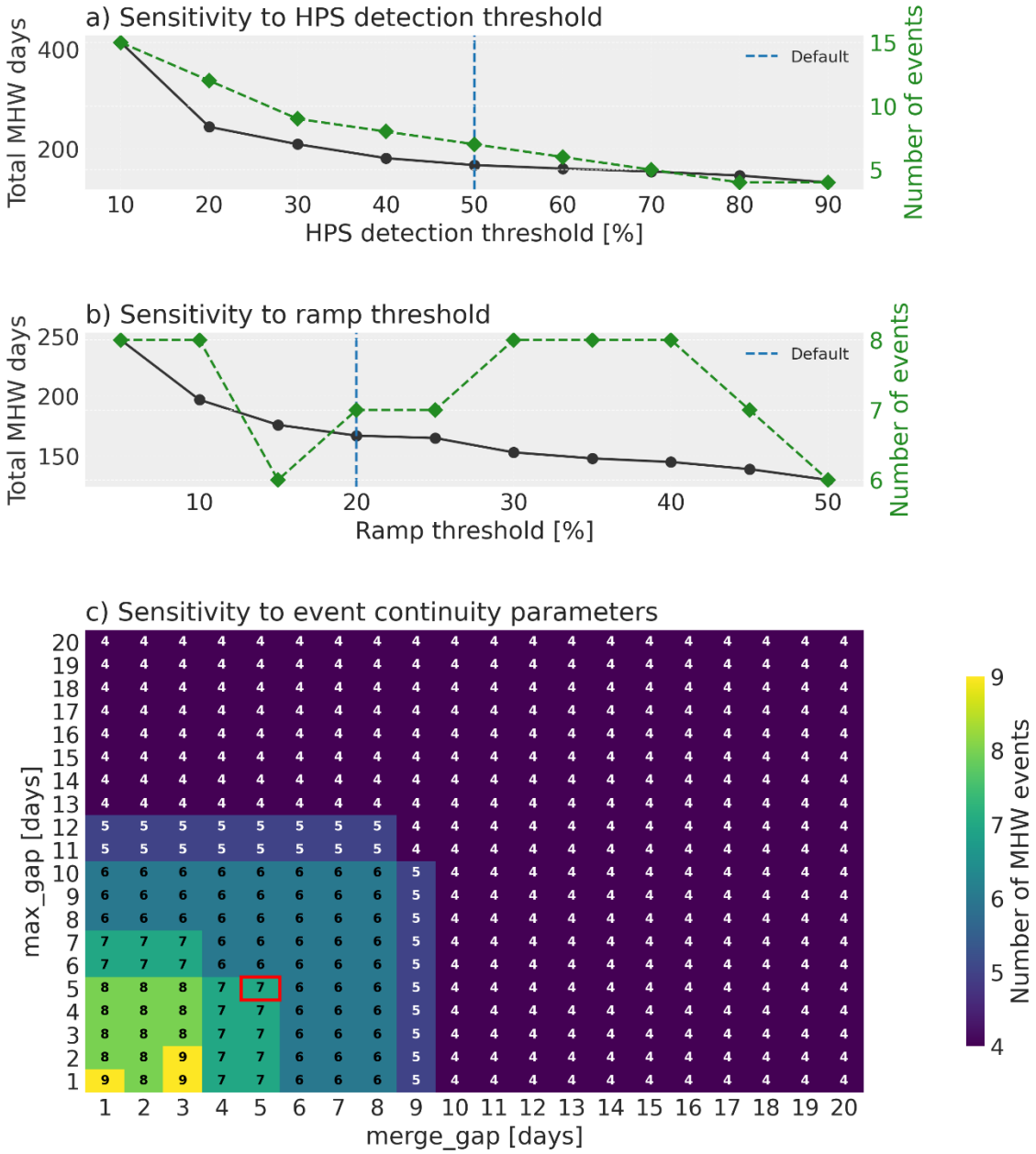


Figure M.2. Sensitivity of HPS-based MHW Event Detection Parameters a) Sensitivity to the HPS detection threshold. Lower thresholds led to an increase in both the number of detected events and the total number of MHW days, reflecting a more conservative detection of events. b) Sensitivity to the ramp threshold. Lower ramp thresholds resulted in longer cumulative MHW duration, whereas higher thresholds reduced the total number of MHW days. The number of detected events showed comparatively weaker sensitivity to this parameter. c) Sensitivity to the event continuity parameters (max_gap and merge_gap), shown as the number of detected events. For sufficiently large values (merge_gap $\gtrsim$ 10 days or max_gap $\gtrsim$ 12 days), the number of detected events stabilised at four, indicating increased merging of adjacent events. In all panels, the blue dashed line (a–b) and red box (c) indicates the default parameter choices used in the main analysis.

Table T1: MCMC Diagnostics Summary. Global model convergence was confirmed across all fitted models, with zero divergences and an R hat statistic strictly below 1.02. Notably, the minimum Tail ESS across all models and parameters was 937. This indicates highly efficient sampling of the posterior distribution tails, providing empirical evidence that 2000 total draws are sufficient for the stable estimation of extreme values, such as the 1st and 99th percentiles.

Site	Subset [years]	Acceptance Rate	Divergence	Max R-hat	Min Tail-ESS	Min Bulk-ESS
South WA	3	0.818	0	1.01	1290.0	2352.0
	5	0.8244	0	1.01	1126.0	2866.0
	7	0.8048	0	1.01	1192.0	2970.0
	12	0.8163	0	1.01	1209.0	2700.0
	30	0.8176	0	1.01	1195.0	2950.0
Japan	3	0.8141	0	1.01	1182.0	2241.0
	5	0.8071	0	1.01	1240.0	2699.0
	7	0.8087	0	1.01	1038.0	2663.0
	12	0.8112	0	1.01	937.0	2391.0
	30	0.8039	0	1.01	1270.0	2128.0
North West Shelf	3	0.8139	0	1.01	1085.0	2081.0
	5	0.8102	0	1.01	978.0	2194.0
	7	0.8101	0	1.01	1307.0	2313.0
	12	0.8114	0	1.01	1285.0	2491.0
	30	0.799	0	1.01	1116.0	2092.0
UK	3	0.8176	0	1.01	1265.0	2539.0
	5	0.8125	0	1.01	1247.0	3005.0
	7	0.8109	0	1.00	1340.0	2813.0
	12	0.8151	0	1.01	1176.0	3066.0
	30	0.8119	0	1.01	1200.0	2094.0
South WA 50m	3	0.8100	0	1.01	1398.0	2547.0
	5	0.8093	0	1.01	1346.0	3116.0
	7	0.8208	0	1.00	1449.0	2836.0
	12	0.8024	0	1.01	1154.0	4220.0
	17	0.8058	0	1.01	1249.0	2687.0

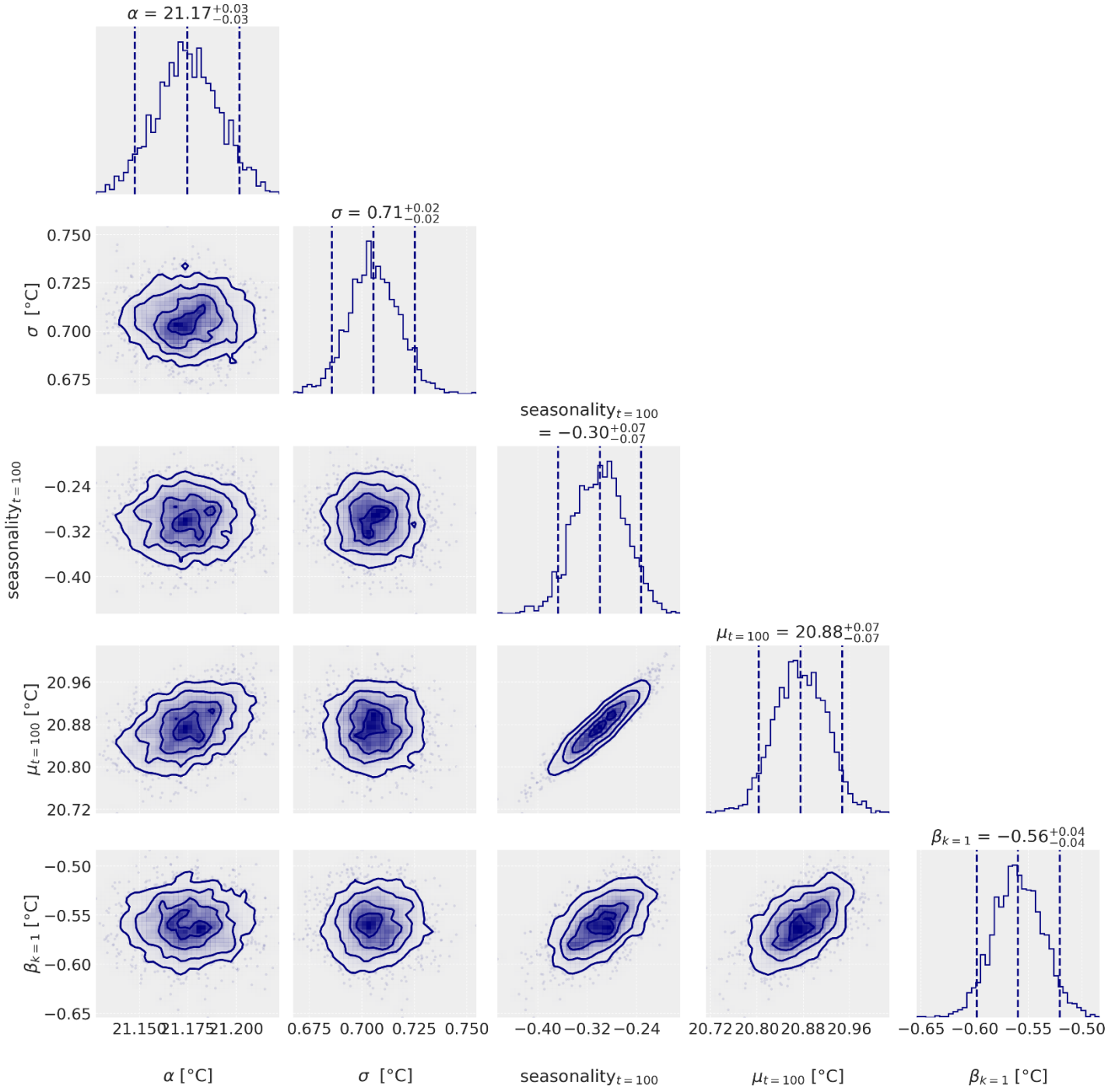


Figure S0.1. Corner plots of the posterior pairwise relationships and marginal distributions of the main model parameters for daily temperature data from a 5-year subset at site South~WA. The parameters are *alpha* and *sigma*, the time-dependent components evaluated at time index *t = 100 seasonality* and *mu*, and the first Fourier coefficient *beta_k=1*. Each diagonal panel displays the marginal posterior distribution for a single parameter, while the off-diagonal panels show the joint posterior density contours between parameter pairs.

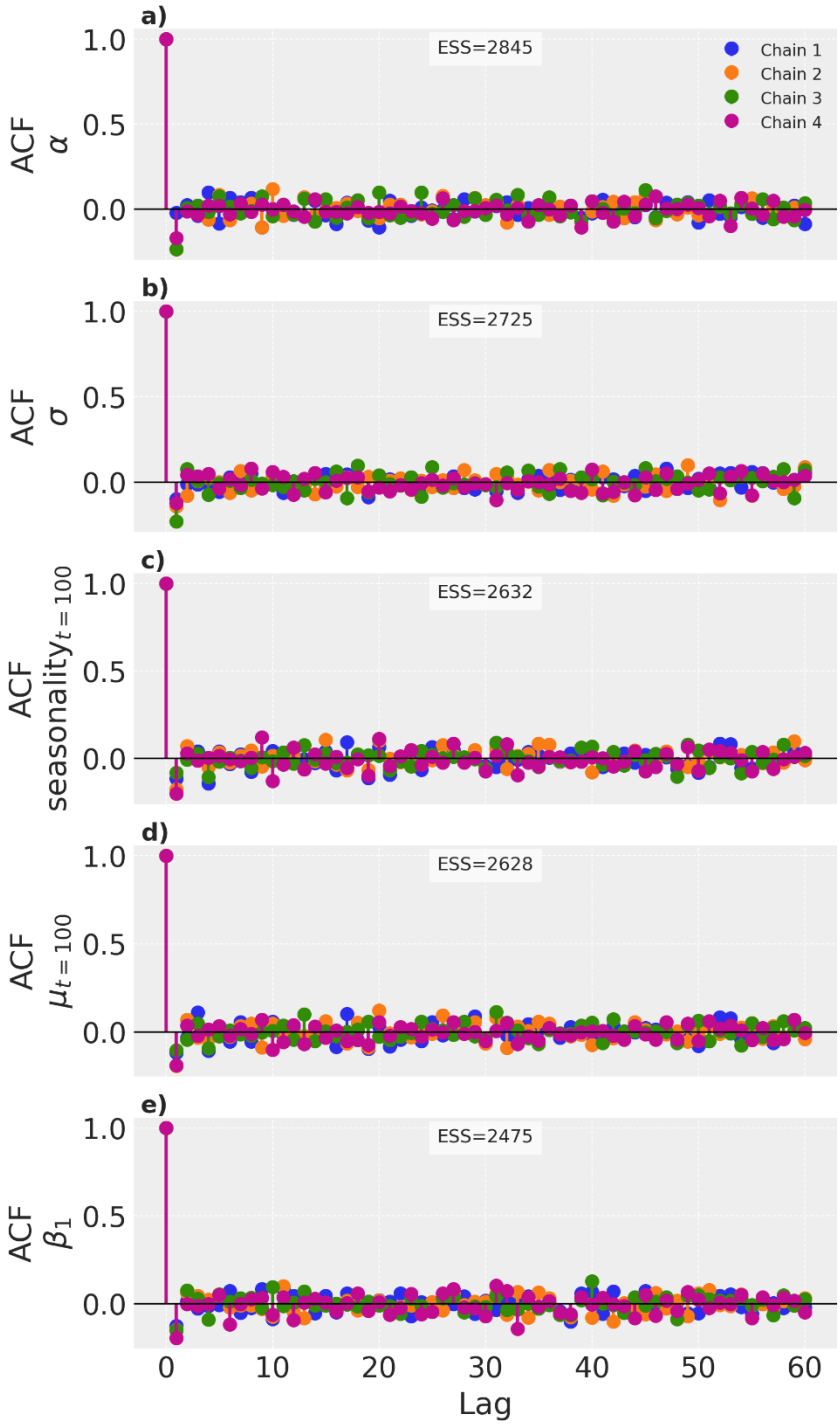


Figure S0.2. Autocorrelation functions (ACF) for model parameters from the 5-year dataset at site South WA. Each panel shows the ACF for all four MCMC chains (chain 1 in blue, chain 2 in orange, chain 3 in green, chain 4 in pink) over 60 lags and the ESS. Panels correspond to: a) *alpha*, b) *sigma*, c) *seasonality_t=100*, d) *mu_t=100*, and e) *beta_k=1*. At lag 0, all chains have ACF = 1, so markers for different chains overlap.

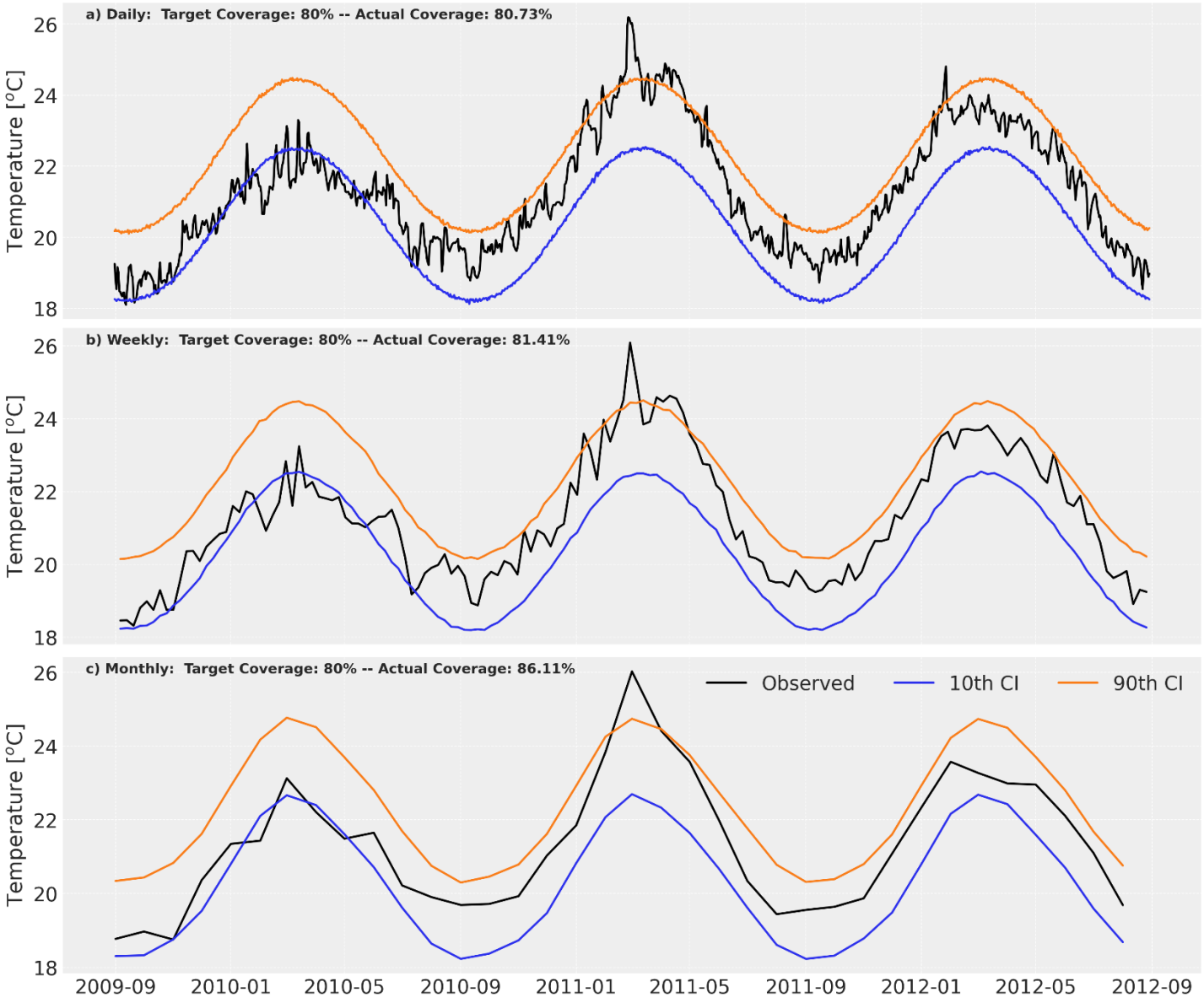


Figure S0.3. Example posterior predictive calibration for a 3-year observational subset at the South WA site. The plots display the actual observed temperature against the model's 10th and 90th predictive credible intervals (representing a 80% target coverage) for a) daily, b) weekly, and c) monthly temporal resolutions.

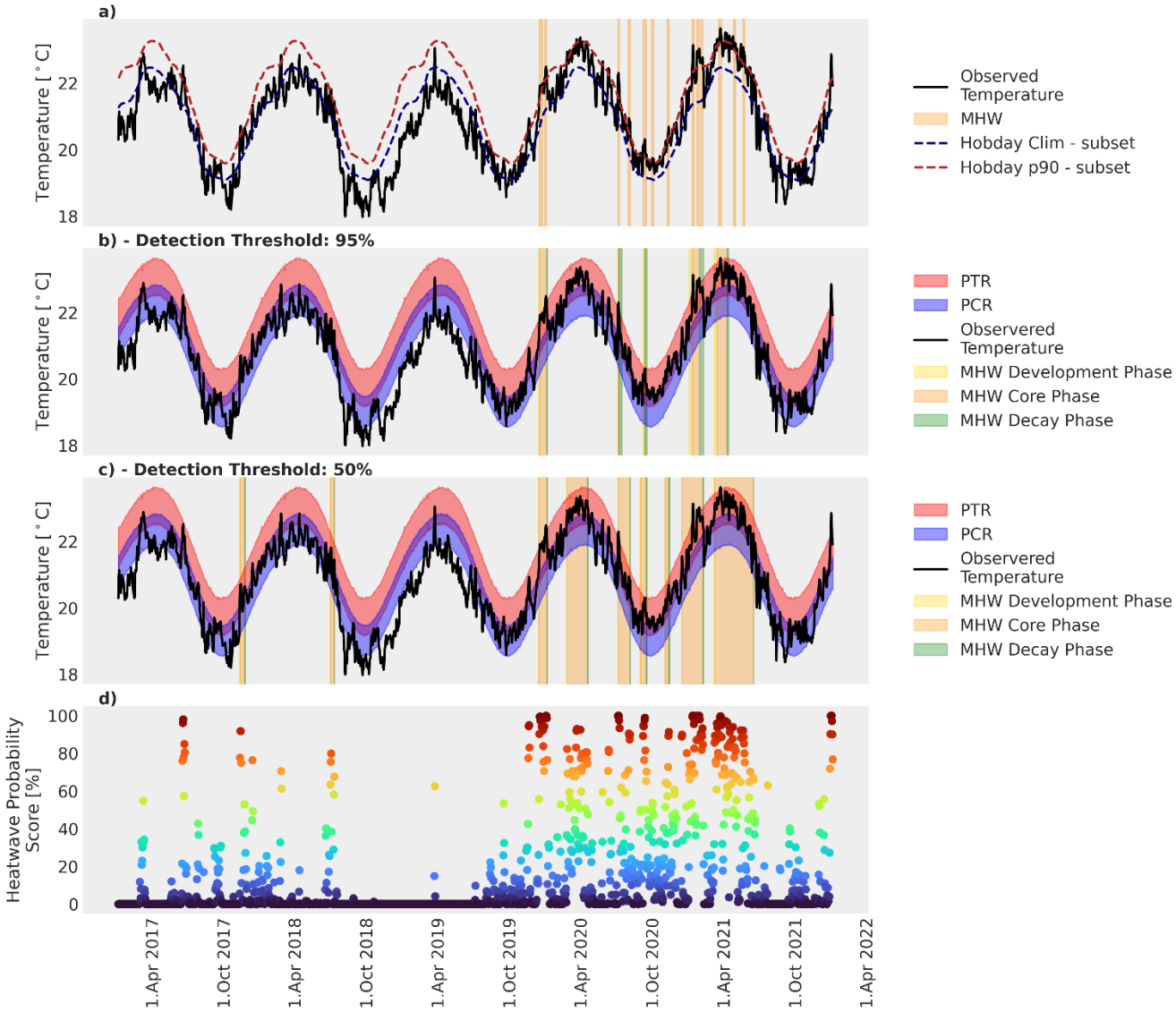


Figure S0.4. Example of probabilistic detection applied to a neutral five year window without major extreme events for Site South WA.

To address the potential bias of selecting subsets centred on known extreme events, we applied the framework to a selected 5 year window (1. January 2017 to 1. January 2022) lacking a documented historical extreme.

As expected under any relative percentile based framework, the statistical definition of a threshold dictates that a fixed proportion of observations will always exceed the baseline. The 90th percentile threshold mathematically ensures that the upper 10 percent of the data will be classified as anomalous. Consequently, applying the model to a neutral period inherently results in a narrower posterior variance and lower inferred thresholds. Standard positive temperature fluctuations are therefore probabilistically categorised as extreme relative to that specific short baseline, guaranteeing the detection of relative events even in the absence of absolute thermal anomalies.

However, if users are aware that a short observational window coincides with a specific large scale climate driver or anomalous regional state, this physical knowledge can be used in the detection threshold of the MHWs. That is a primary advantage of this Bayesian formulation. It has the capacity to integrate broader climatic background states into the MHW detection framework.

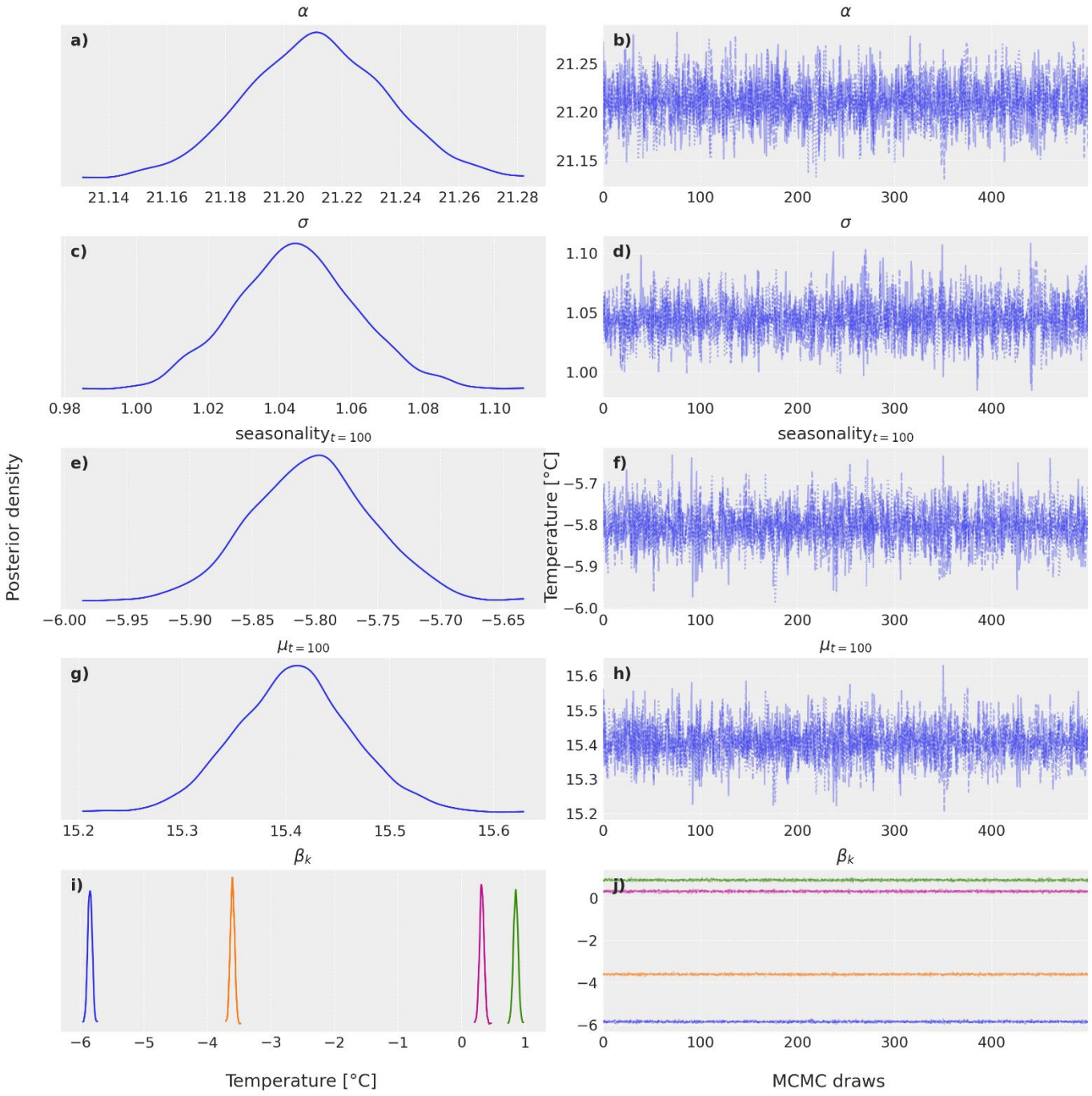


Figure S1.1. Same as Figure 4 but for site Japan**.**

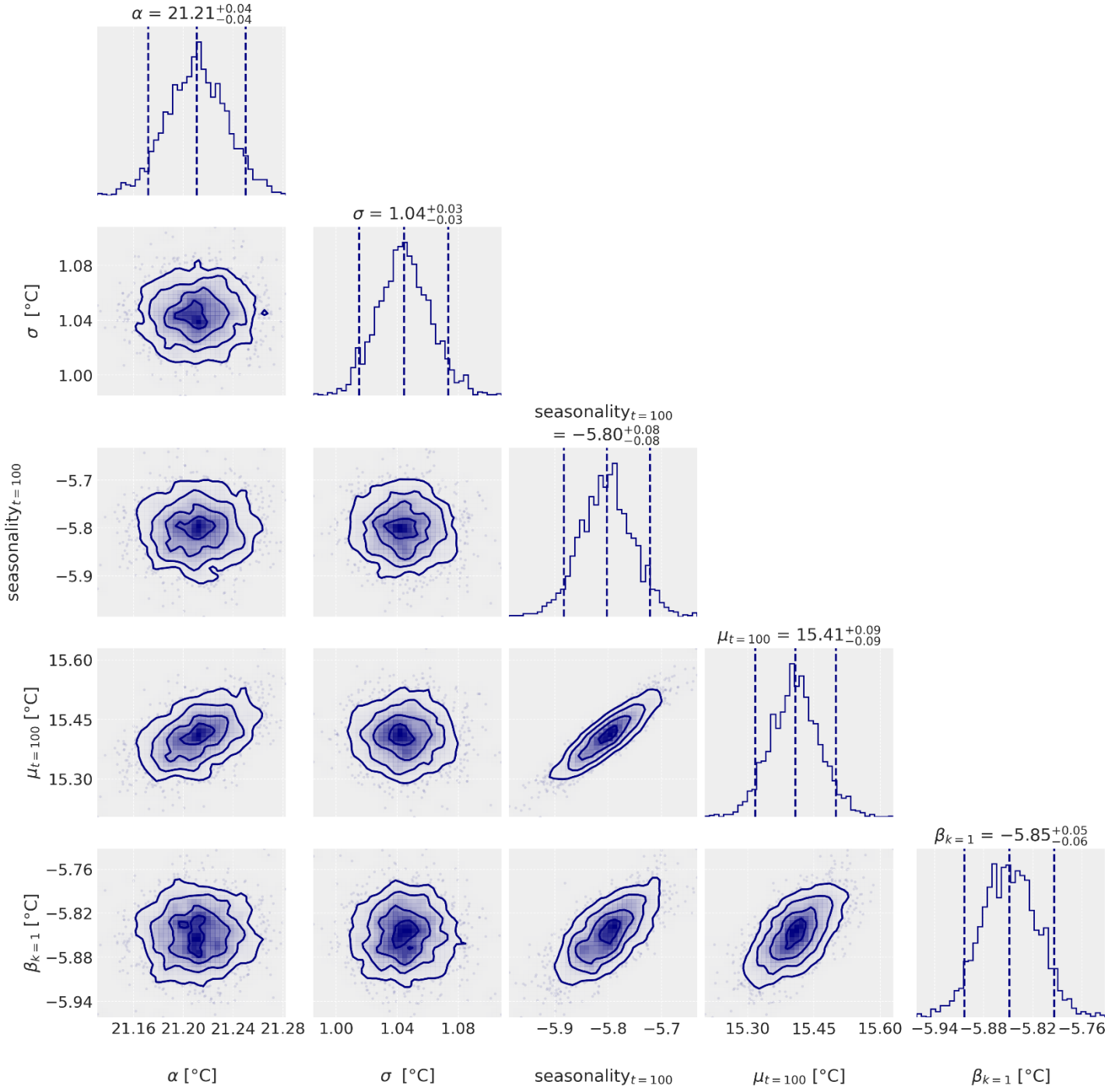


Figure S1.2. Same as Figure S0.1 but for site Japan.

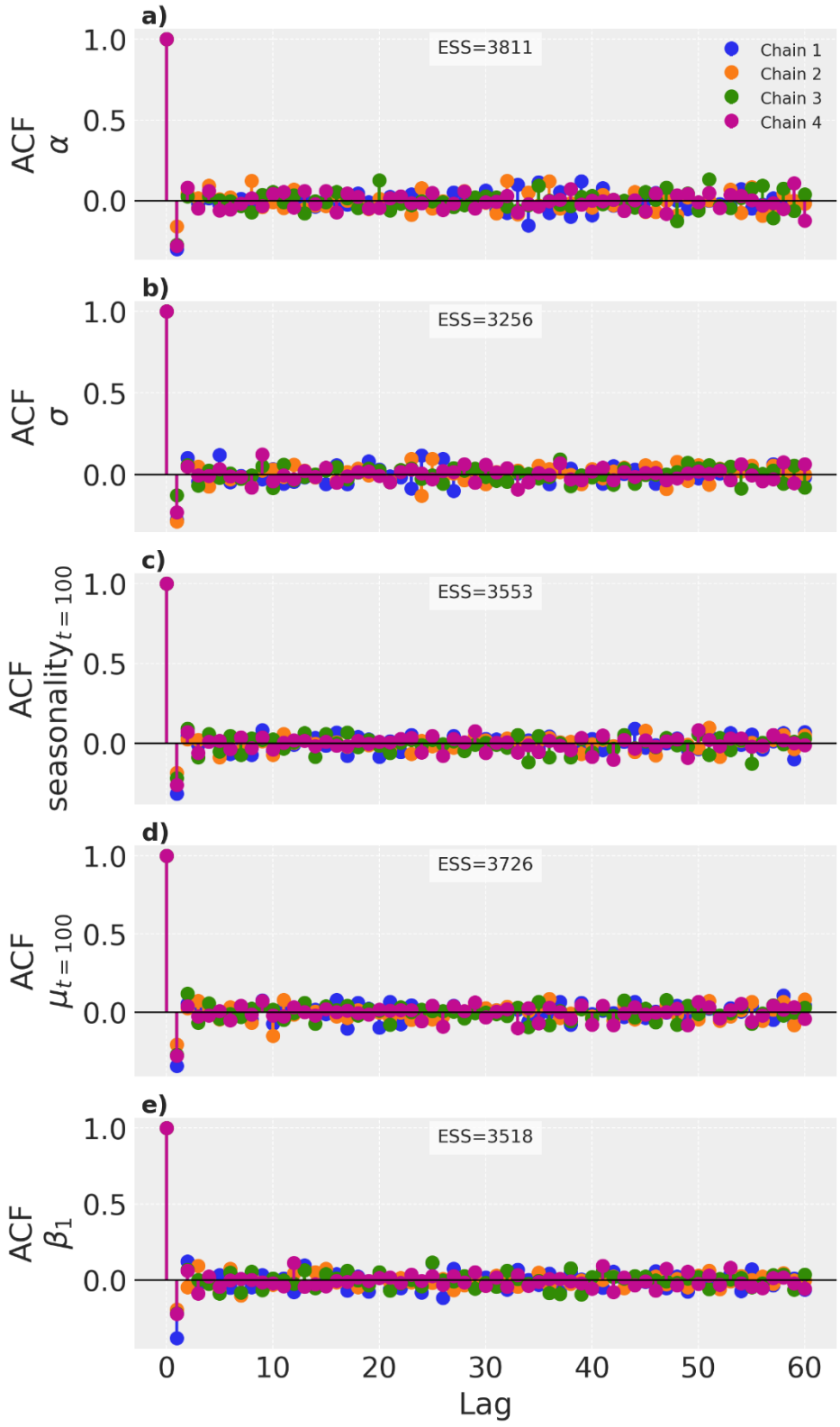


Figure S1.3. Same as Figure S0.2 but for site Japan.

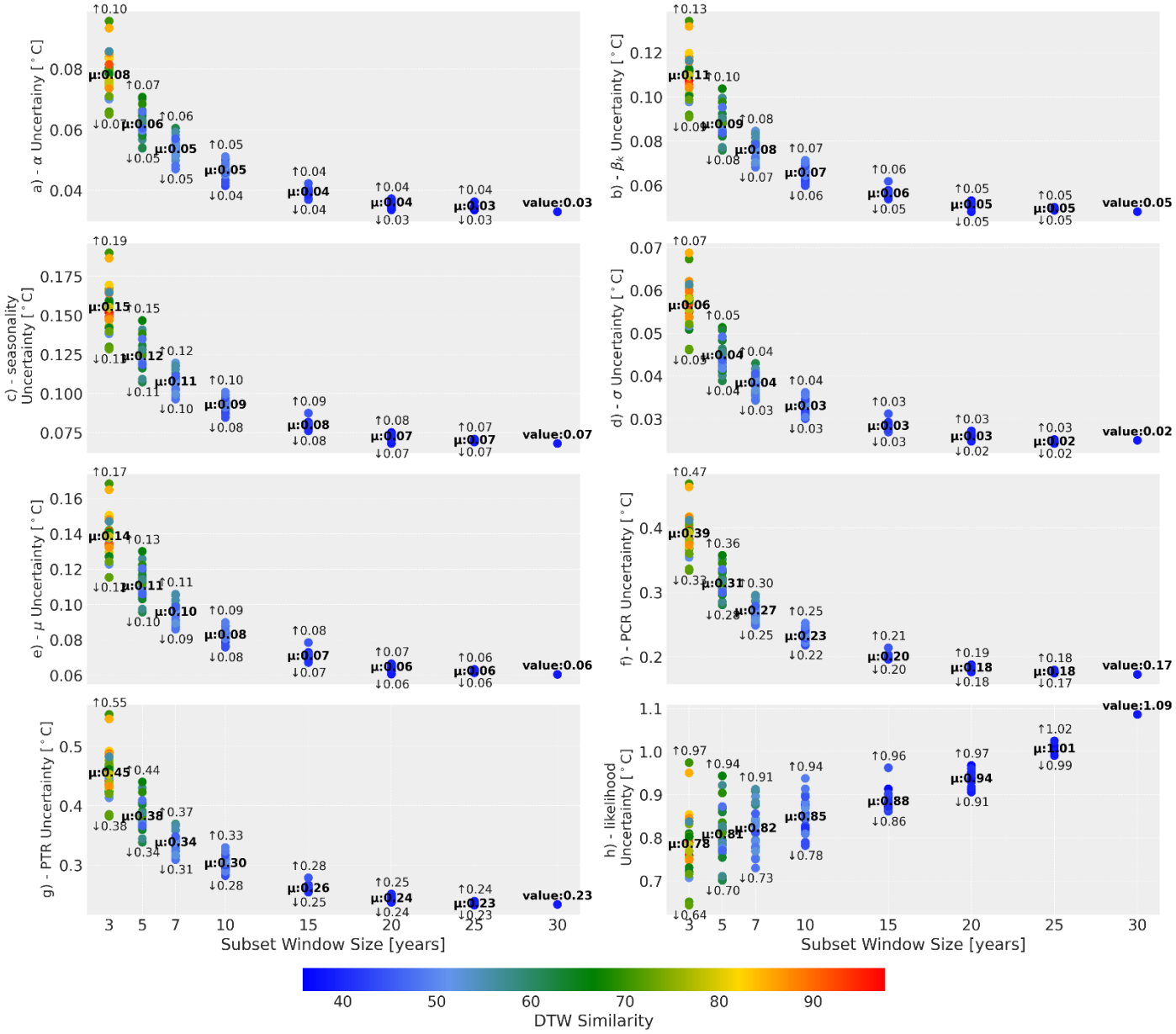


Figure S1.4. Same as Figure 5 but for site Japan.

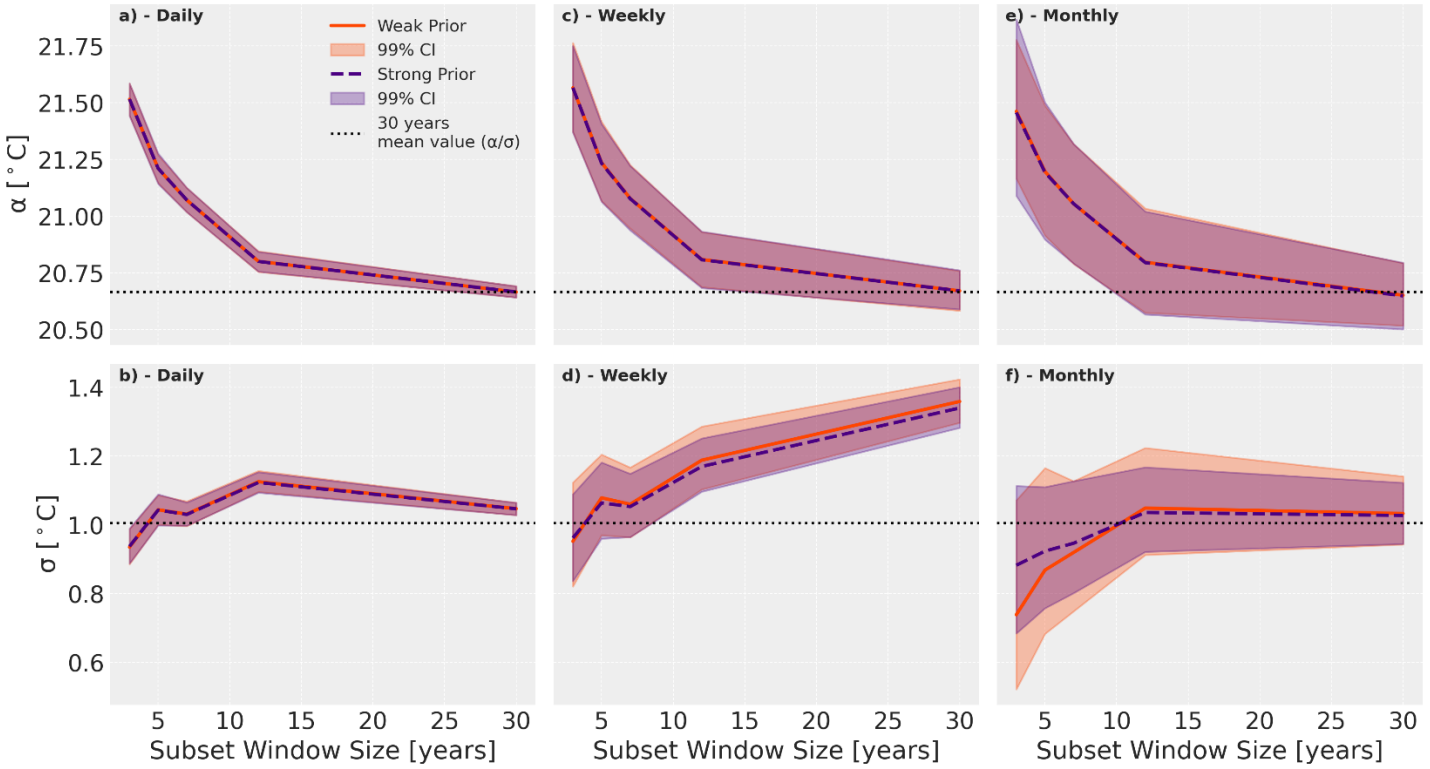


Figure S1.5. Same as Figure 6 but for site Japan.

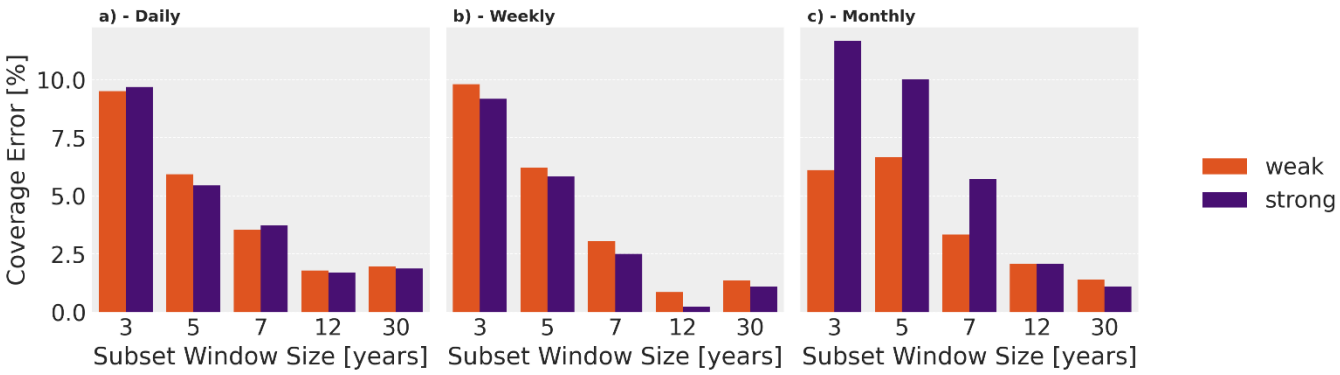


Figure S1.6. Same as Figure 7 but for site Japan.

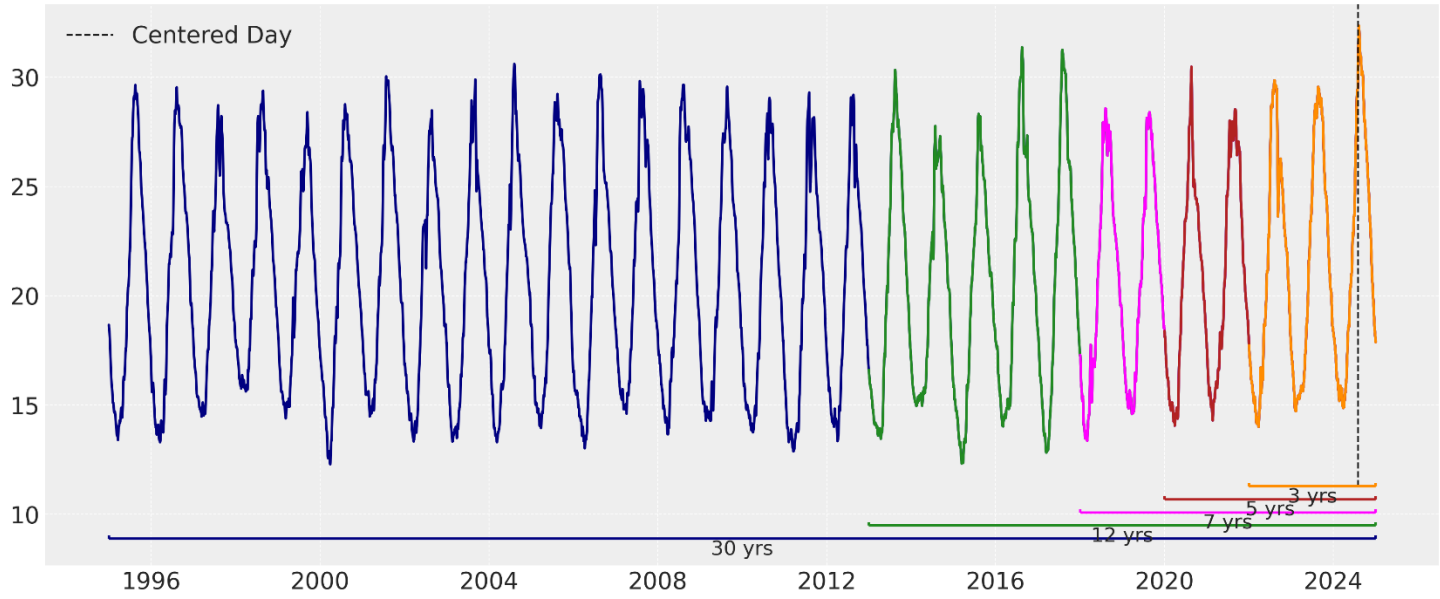


Figure S1.7. Same as Figure 8 but for site Japan.

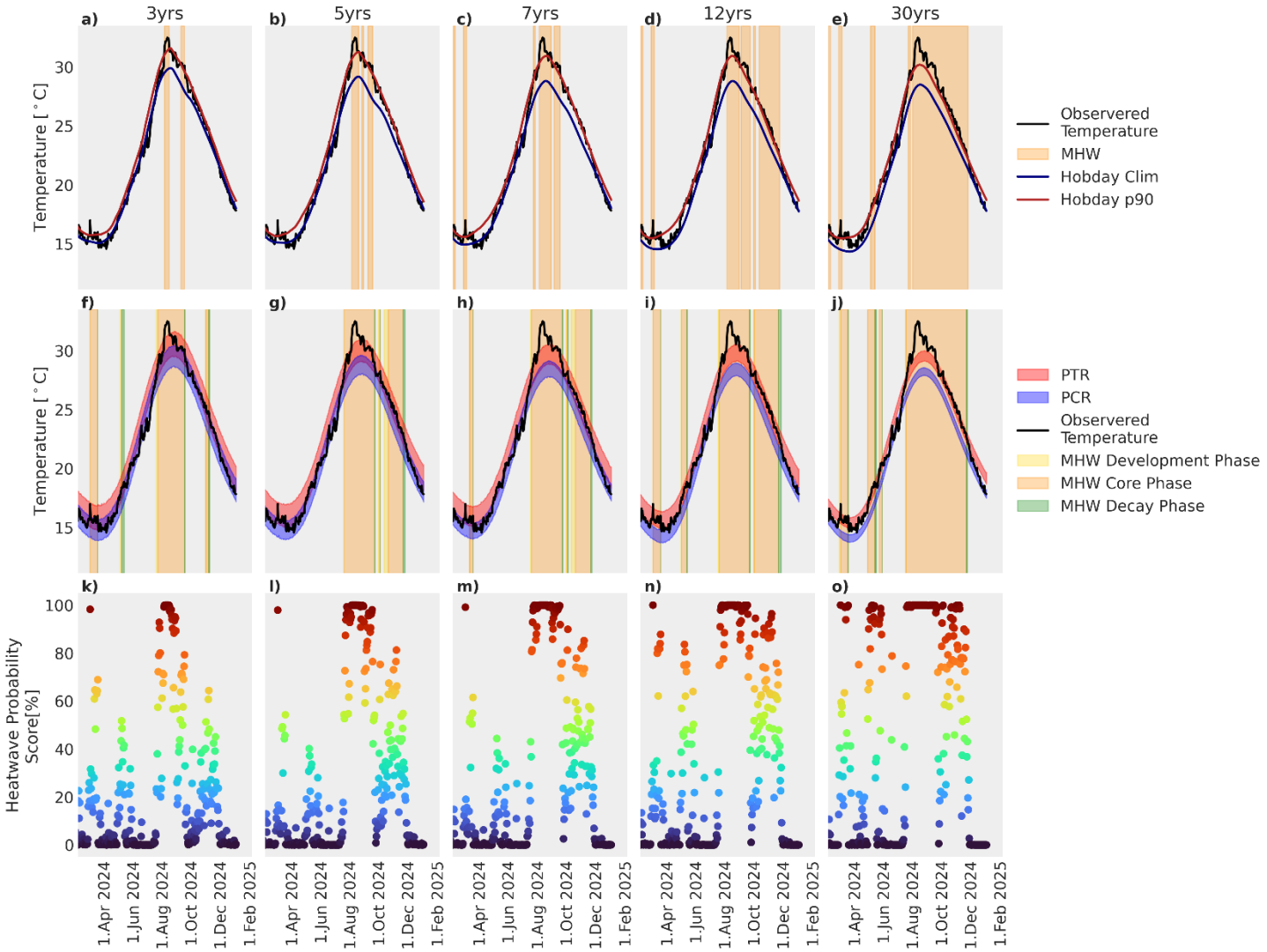


Figure S1.8. Same as Figure 9 but for site Japan.

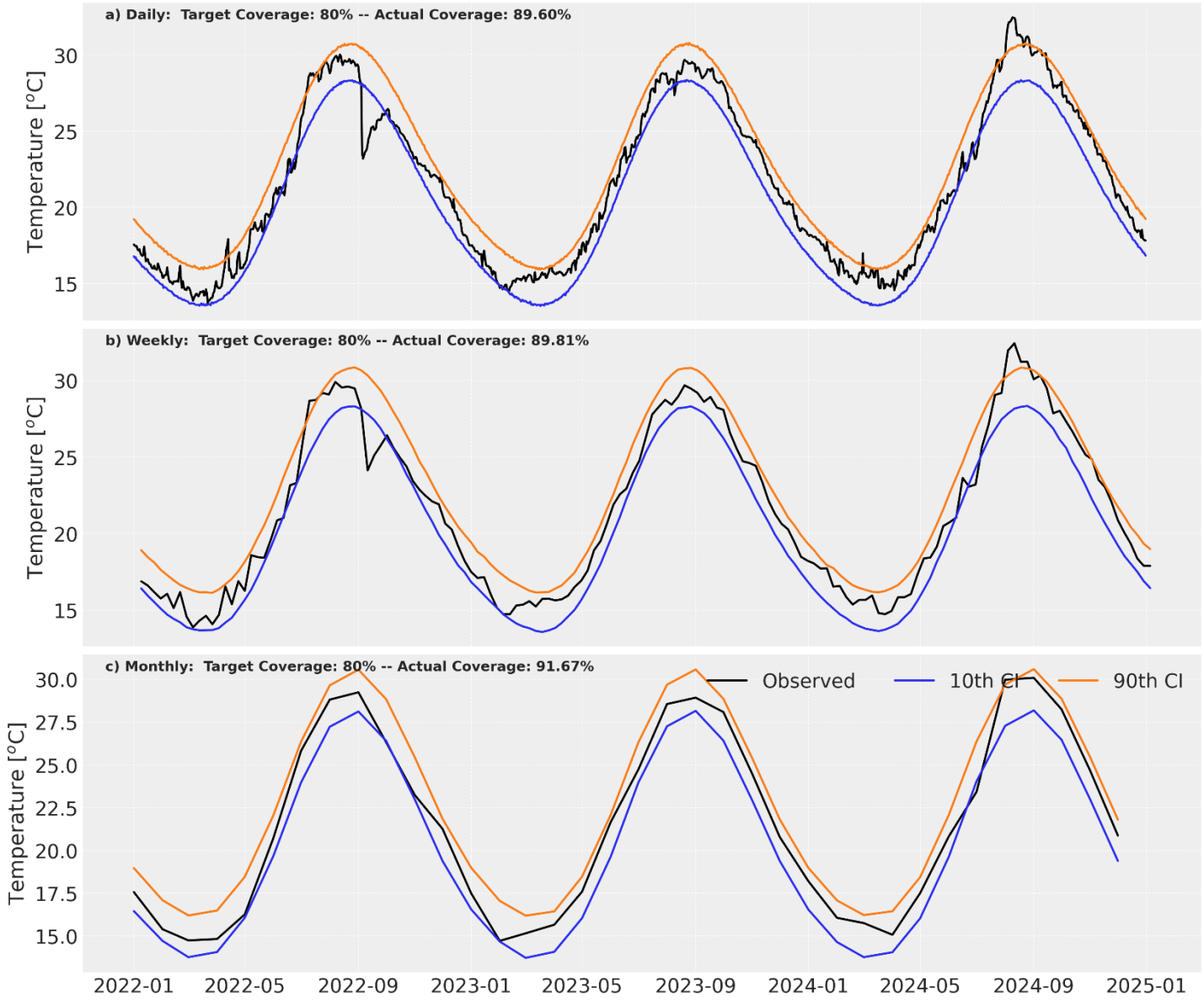


Figure S1.9. Same as Figure S0.3 but for site Japan.

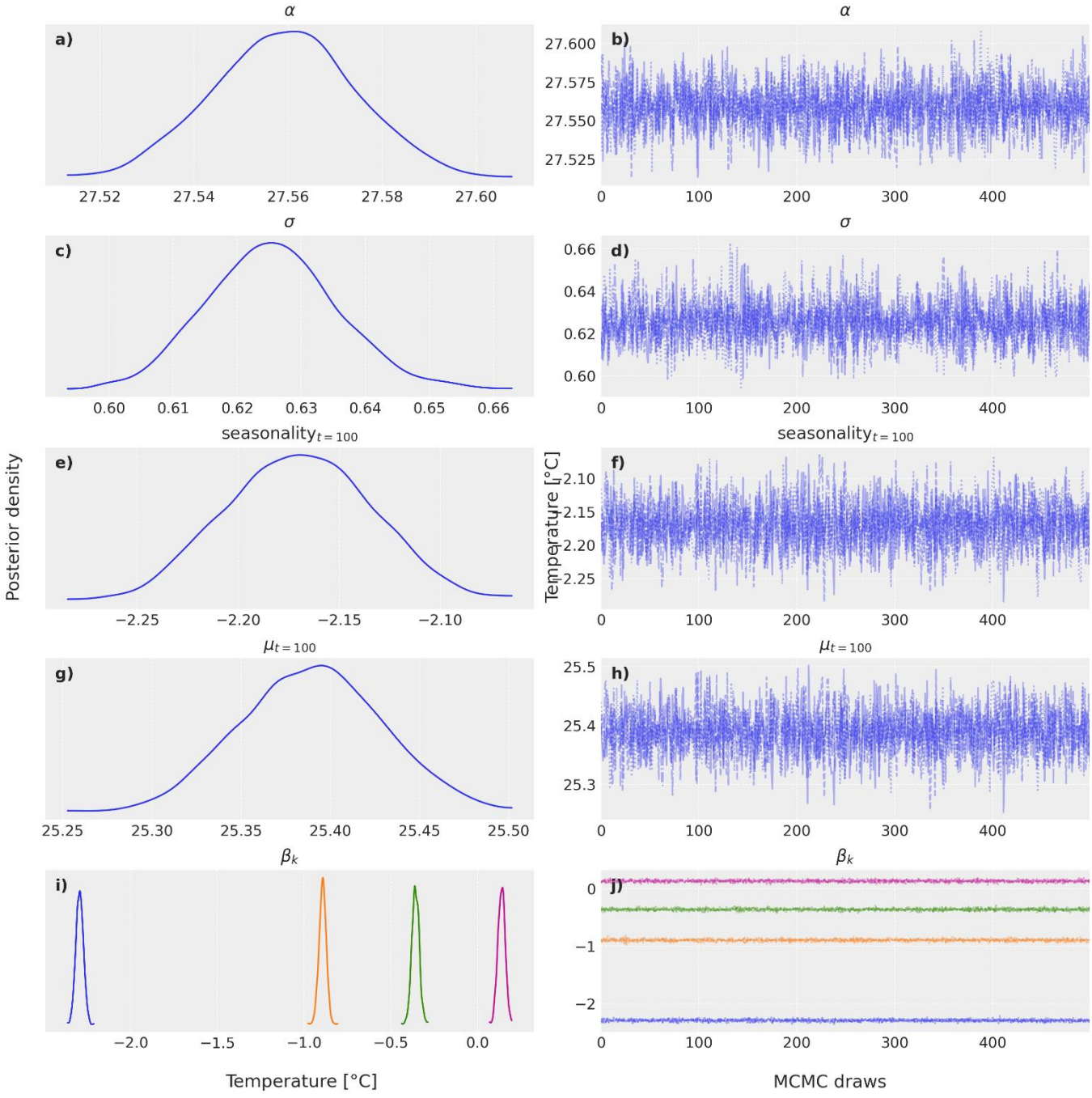


Figure S2.1. Same as Figure 4 but for site North West Shelf**.**

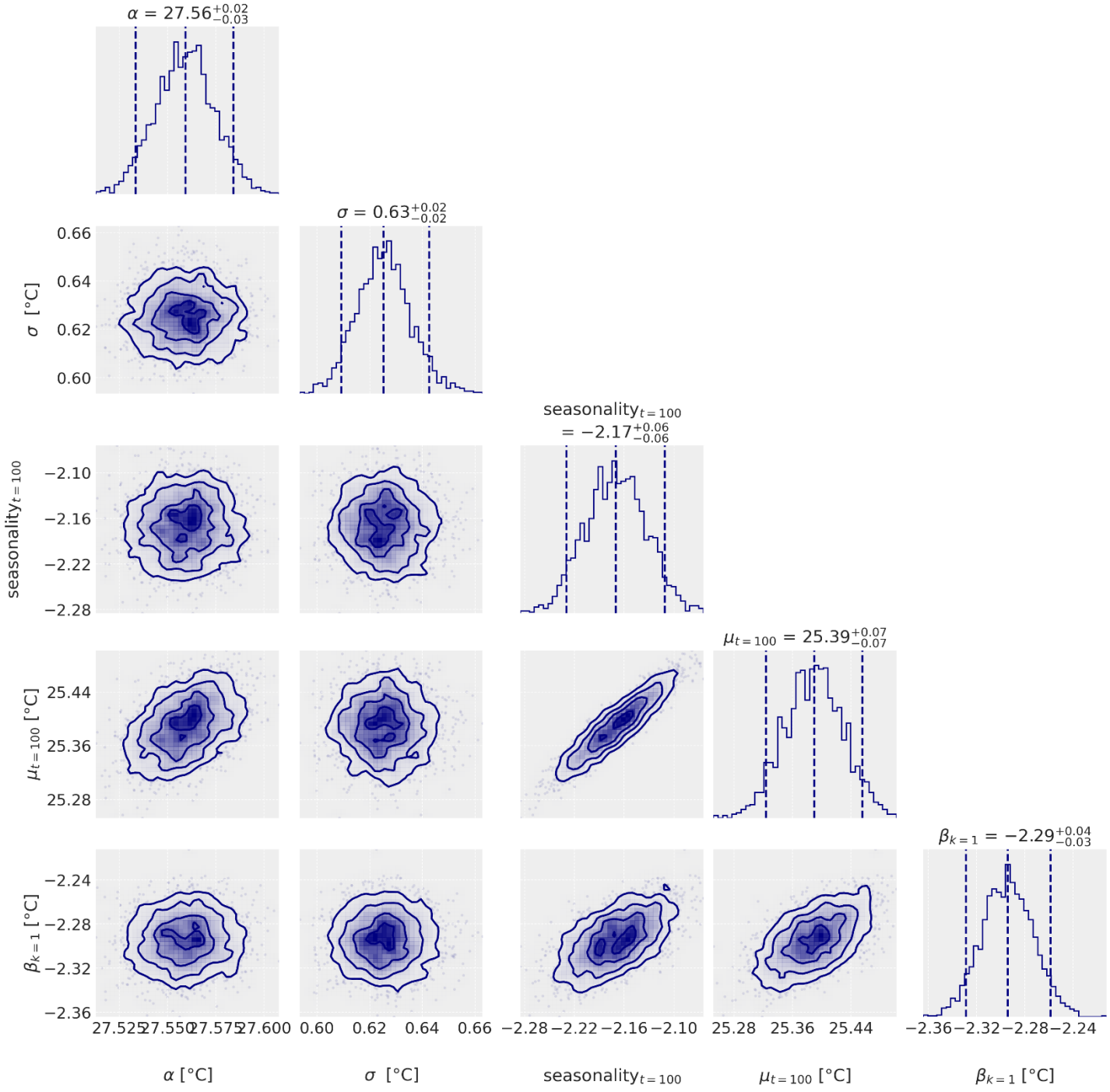


Figure S2.2. Same as Figure S0.1 but for site North West Shelf.

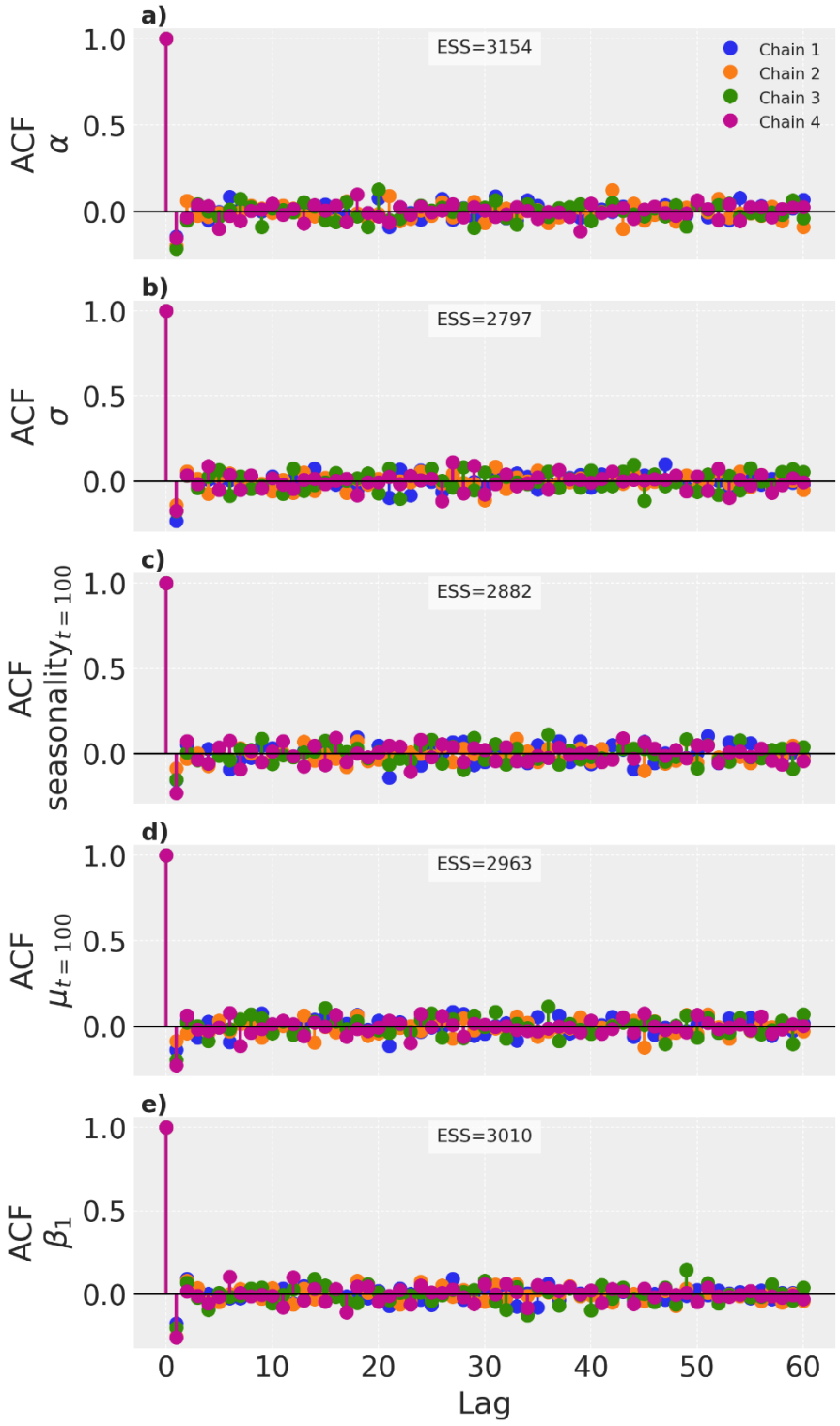


Figure S2.3. Same as Figure S0.2 but for site North West Shelf.

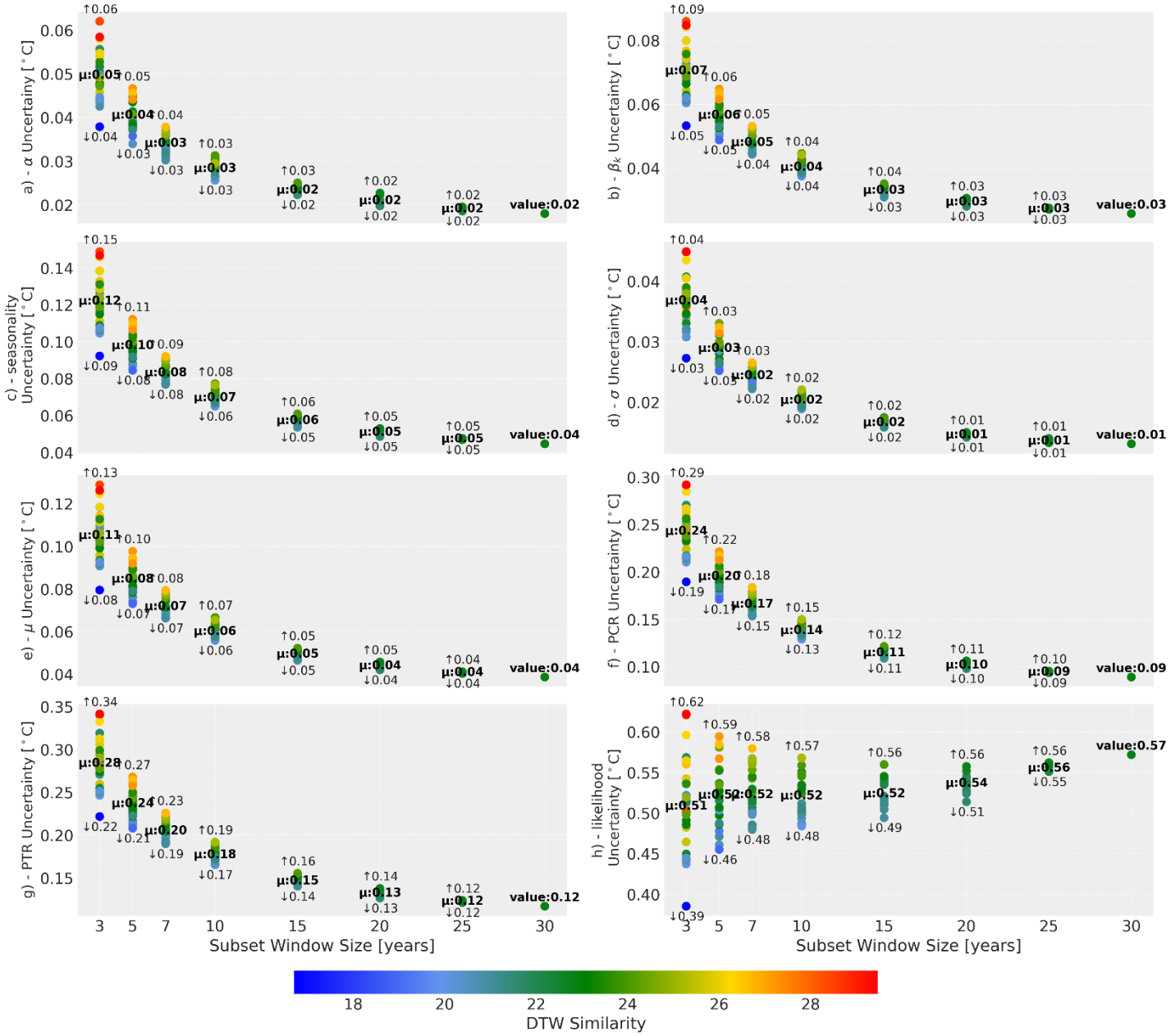


Figure S2.4. Same as Figure 5 but for site North West Shelf.

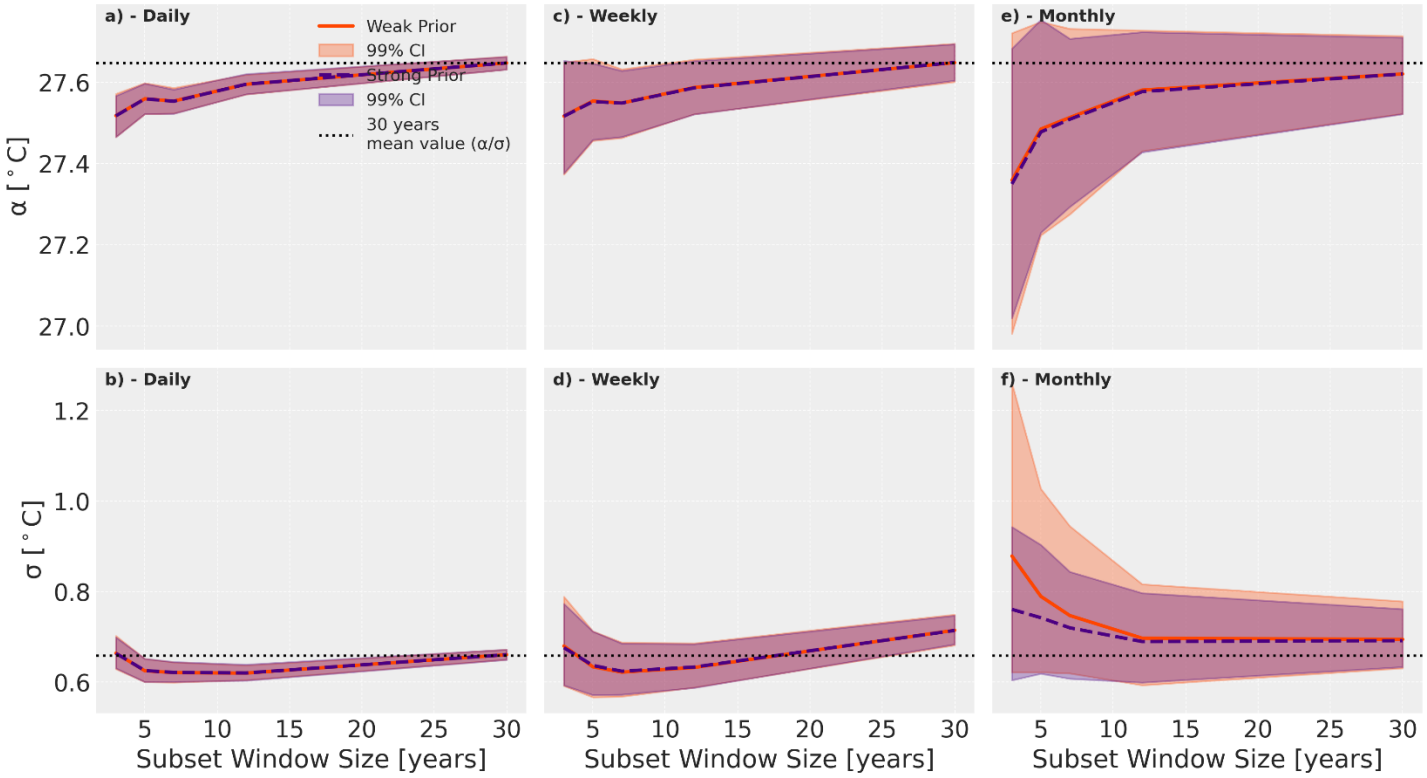


Figure S2.5. Same as Figure 6 but for site North West Shelf.

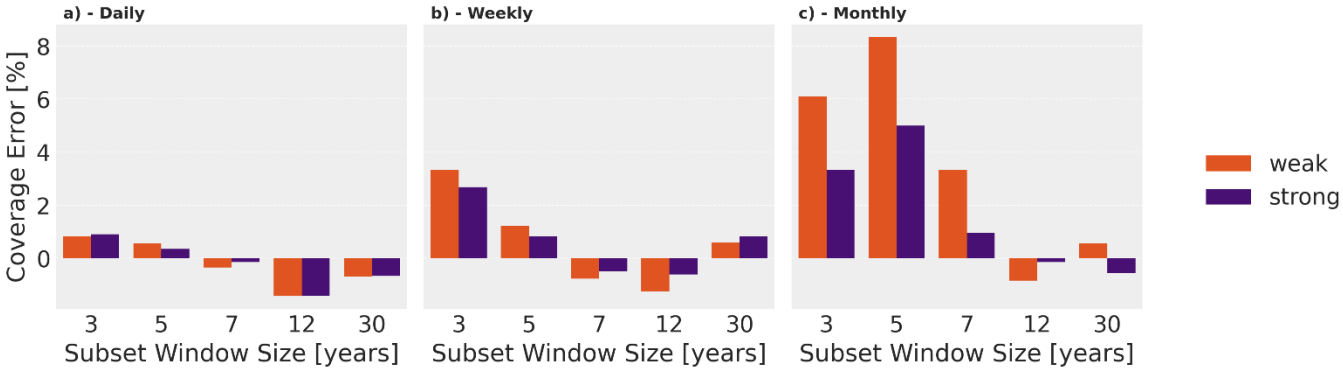


Figure S2.6. Same as Figure 7 but for site North West Shelf.

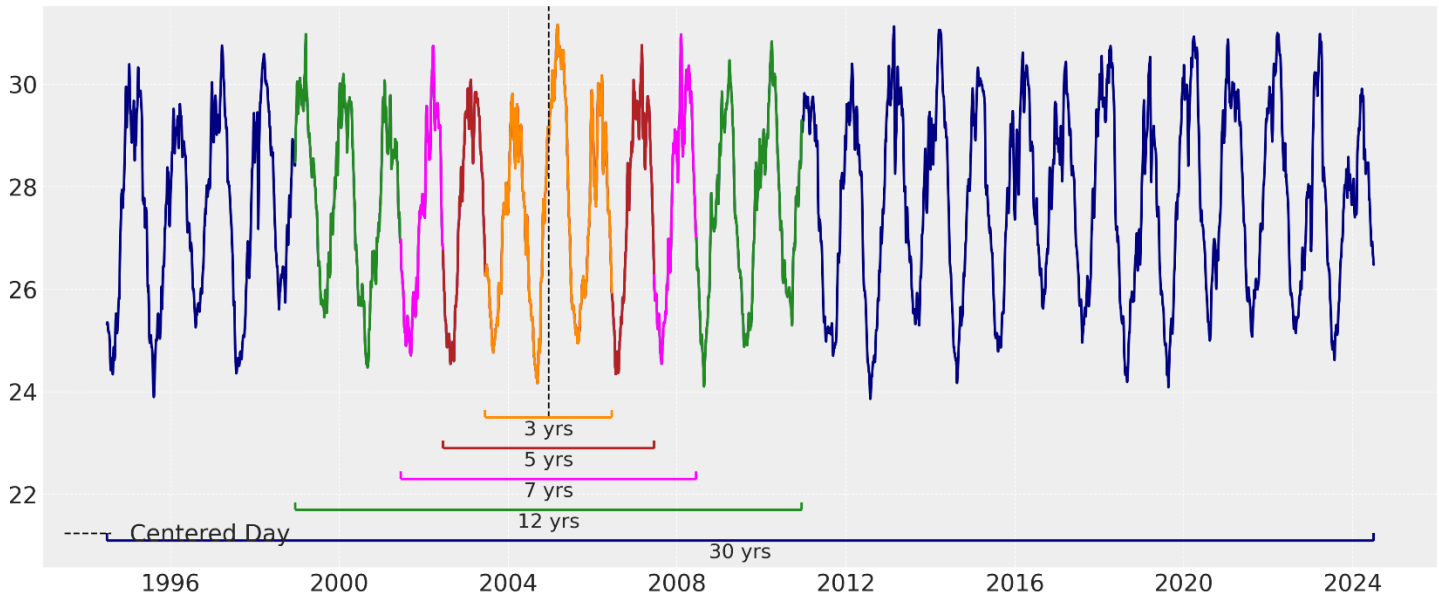


Figure S2.7. Same as Figure 8 but for site North West Shelf.

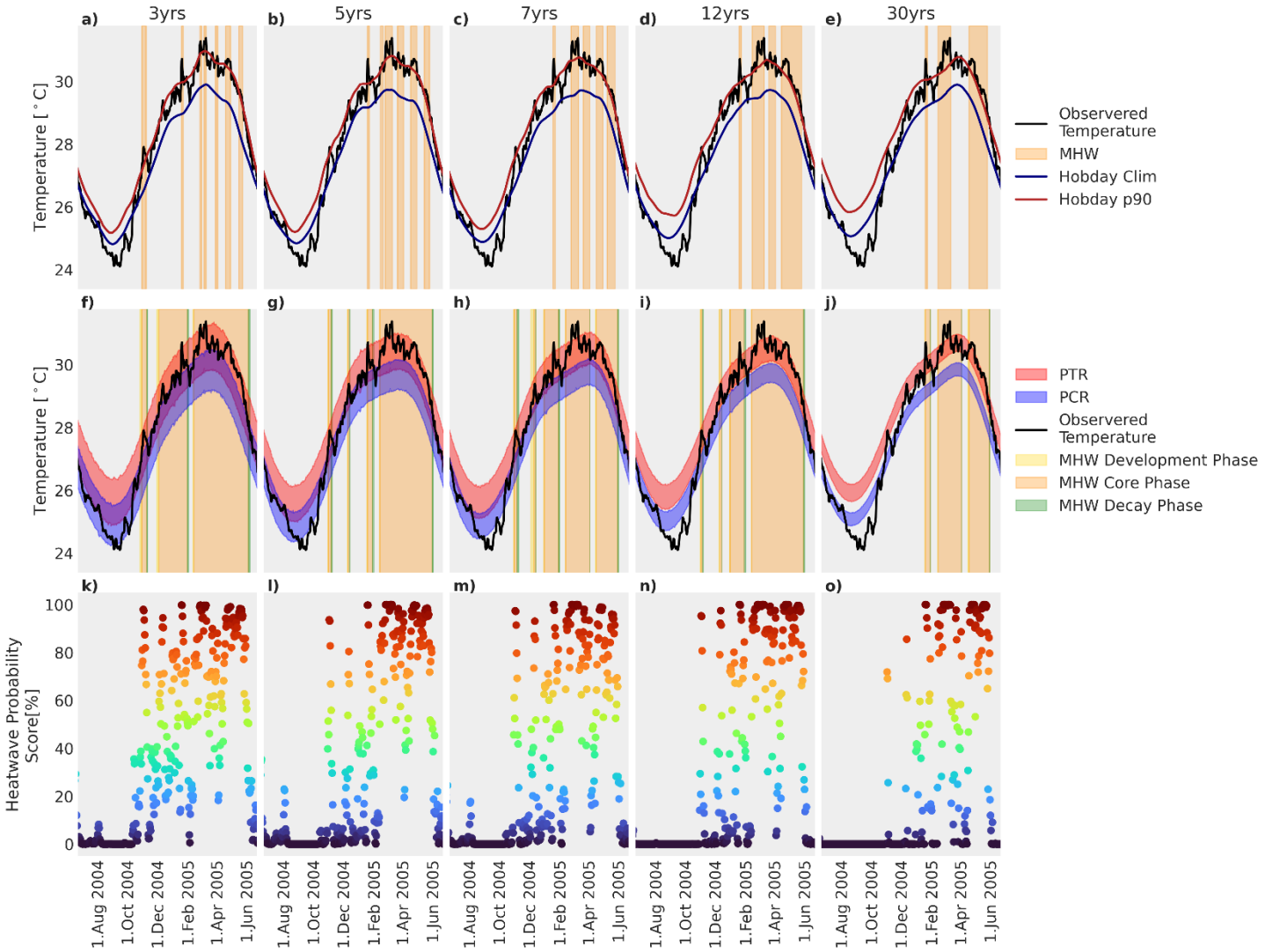


Figure S2.8. Same as Figure 9 but for site North West Shelf.

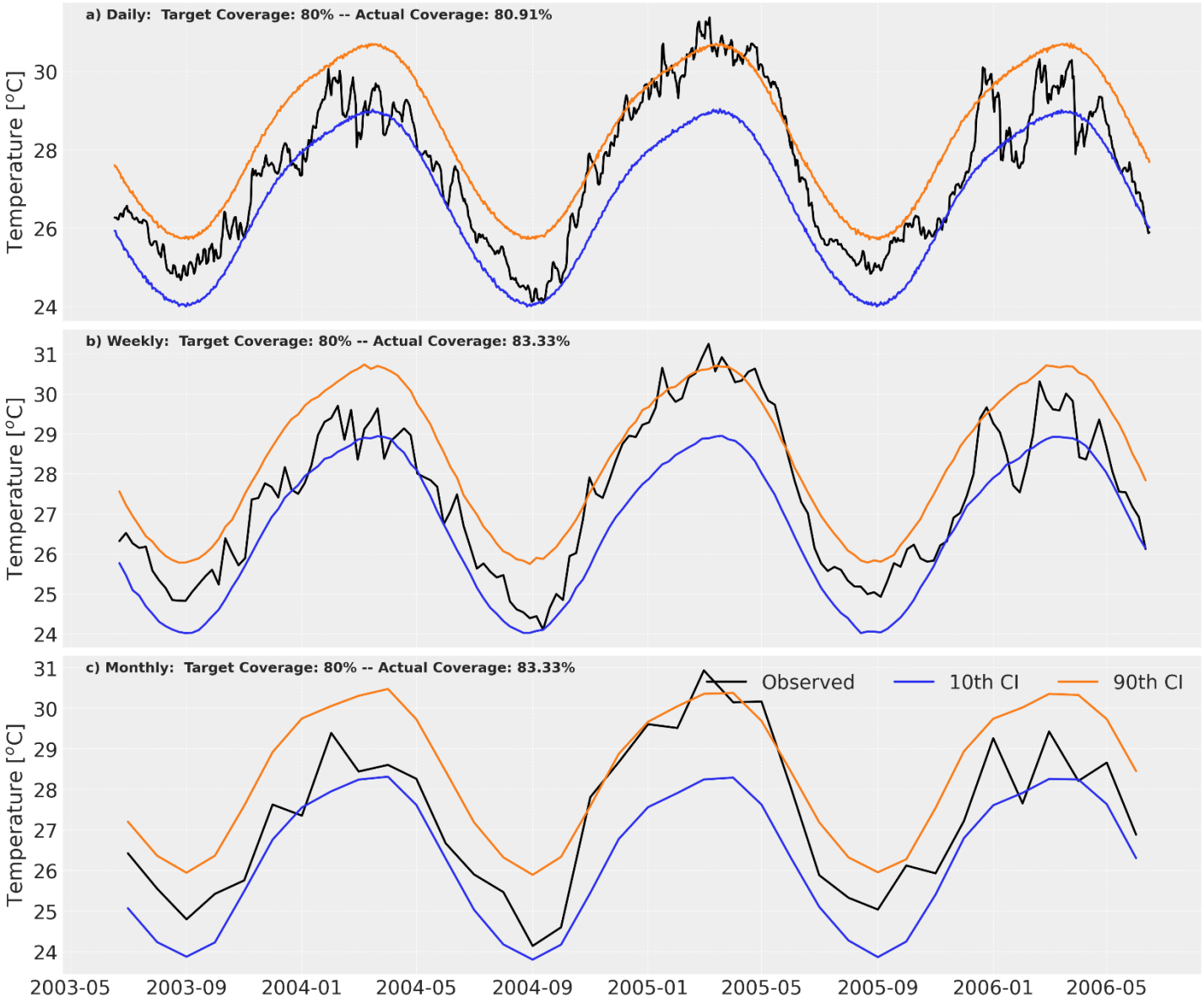


Figure S2.9. Same as Figure S0.3 but for site North West Shelf.

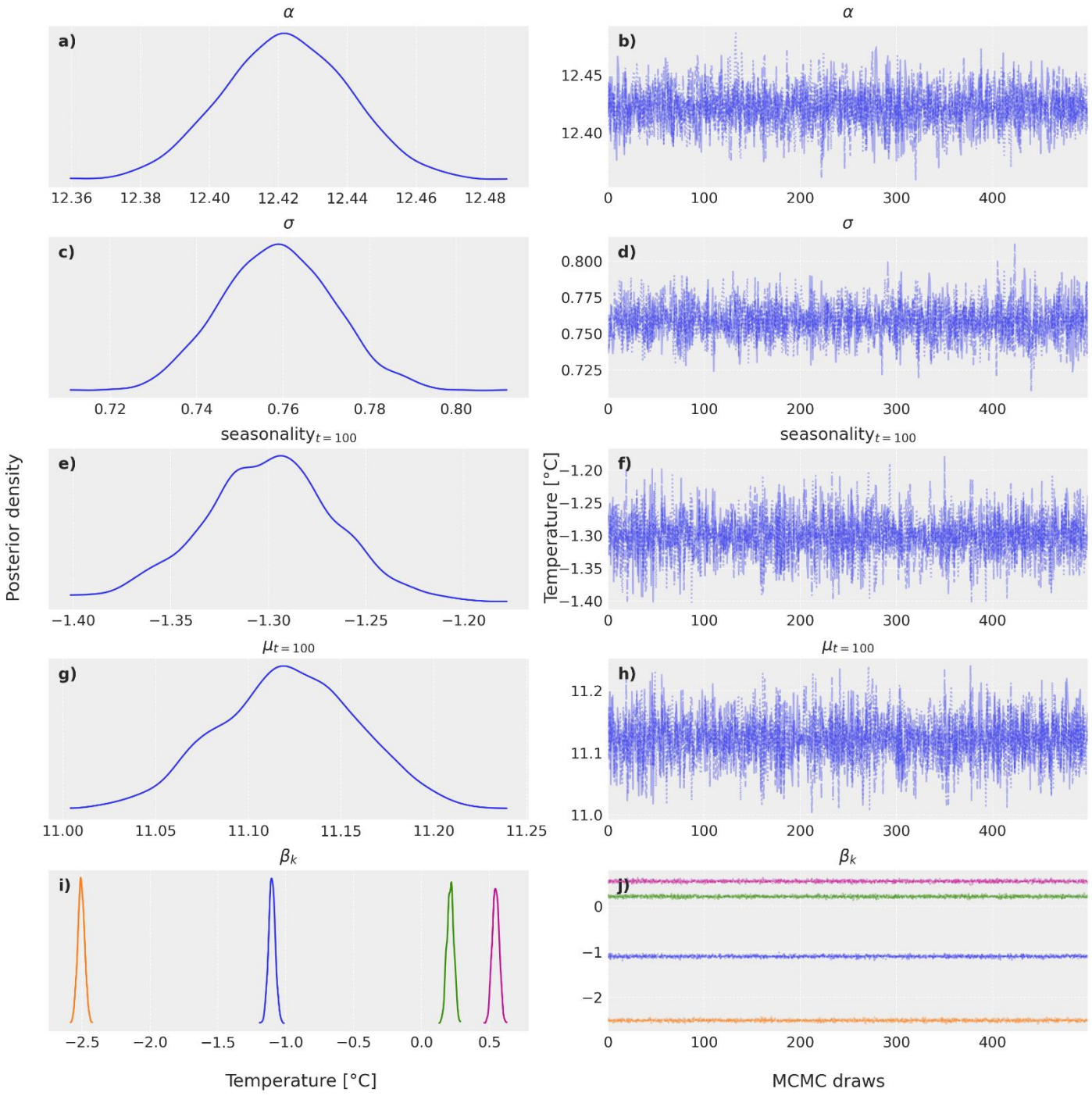


Figure S3.1. Same as Figure 4 but for site UK**.**

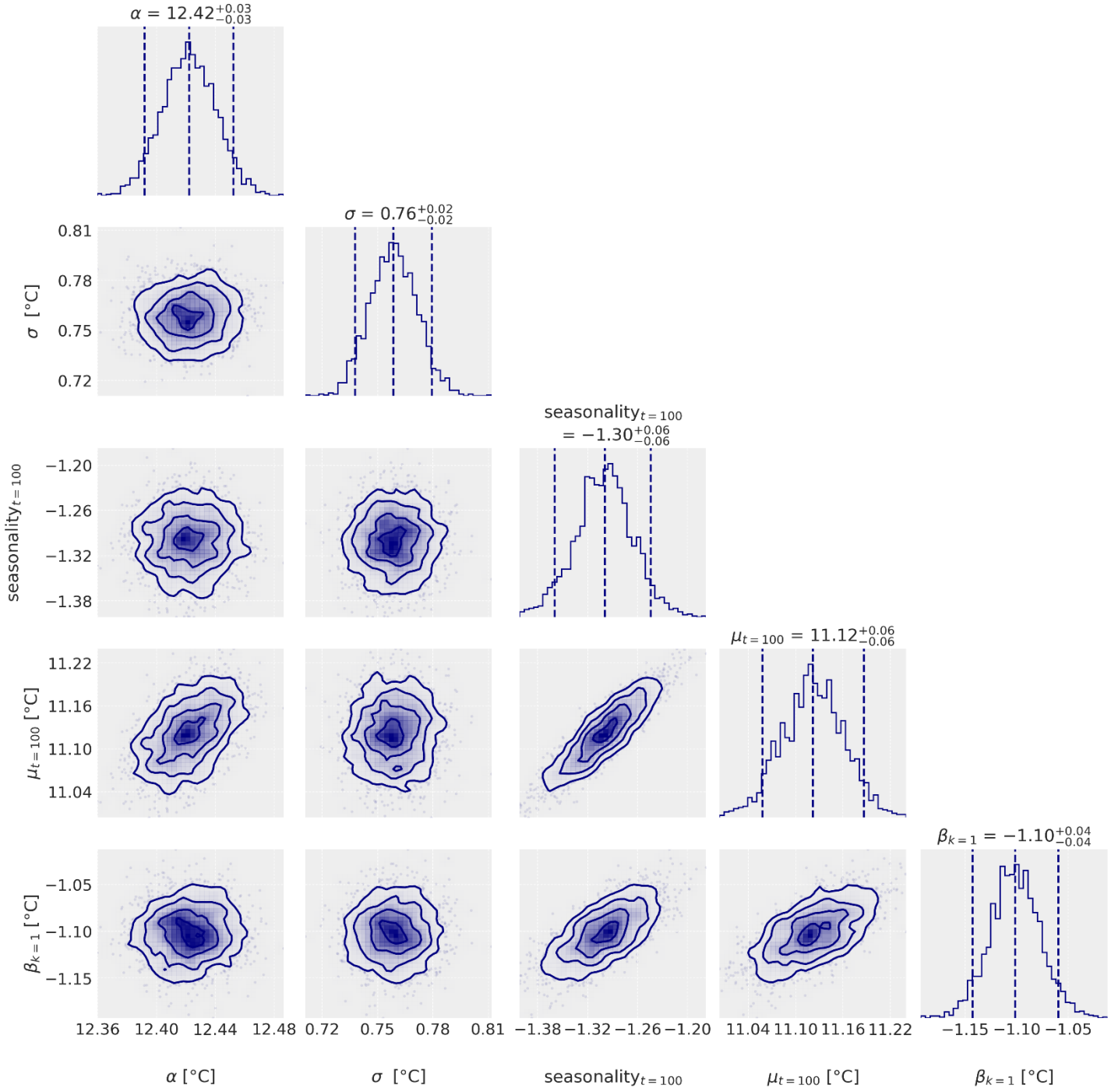


Figure S3.2. Same as Figure S0.1 but for site UK.

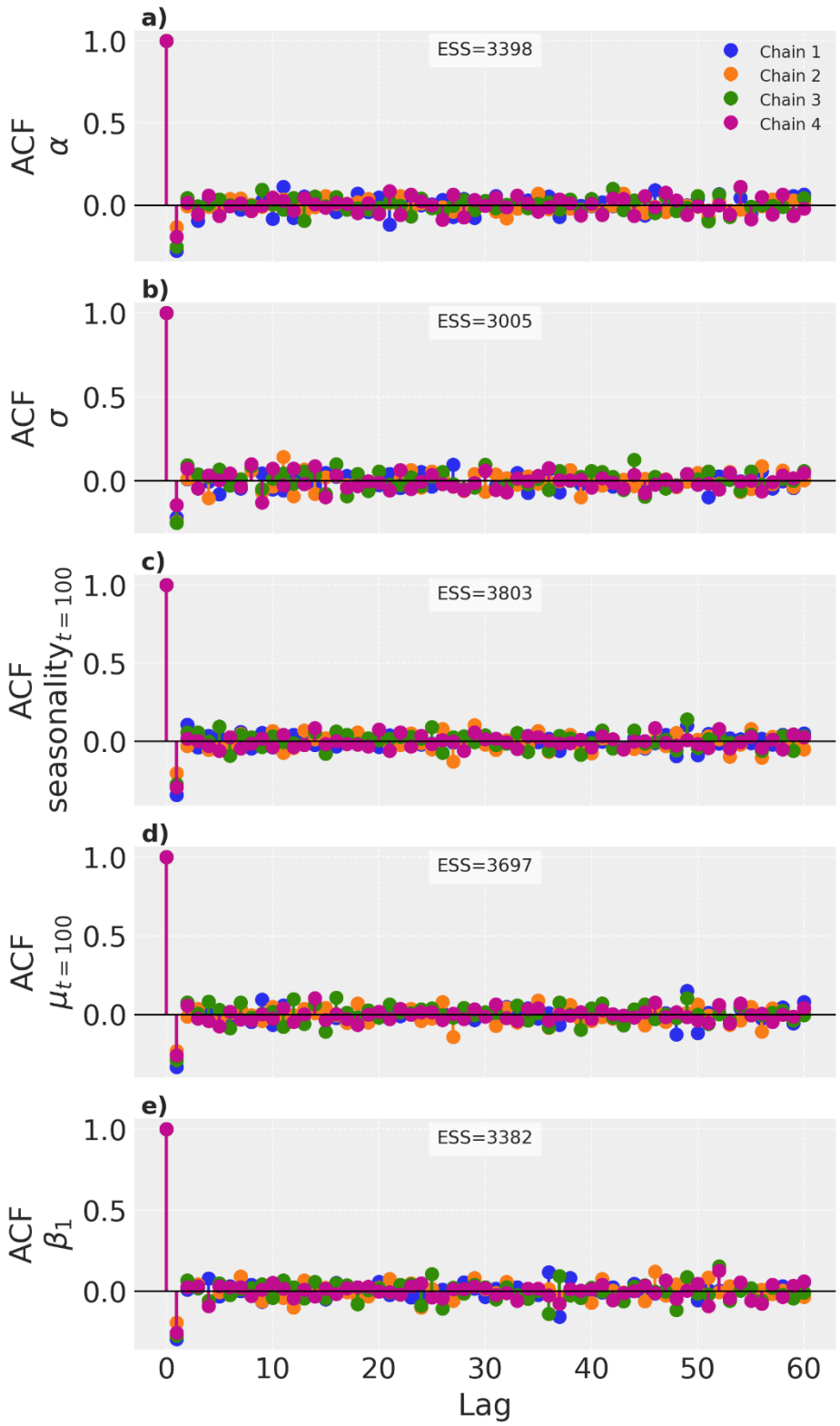


Figure S3.3. Same as Figure S0.2 but for site UK.

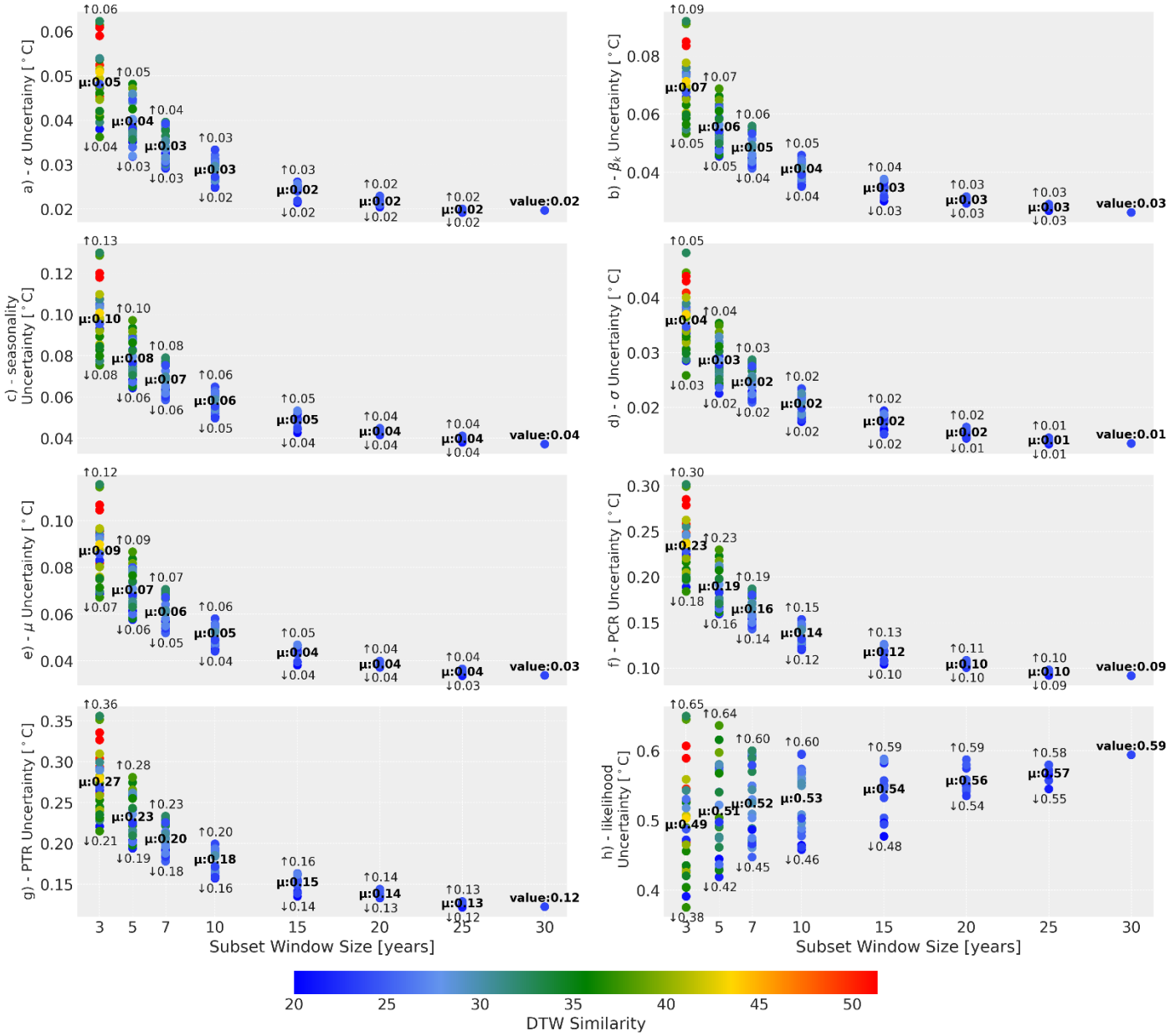


Figure S3.4. Same as Figure 5 but for site UK.

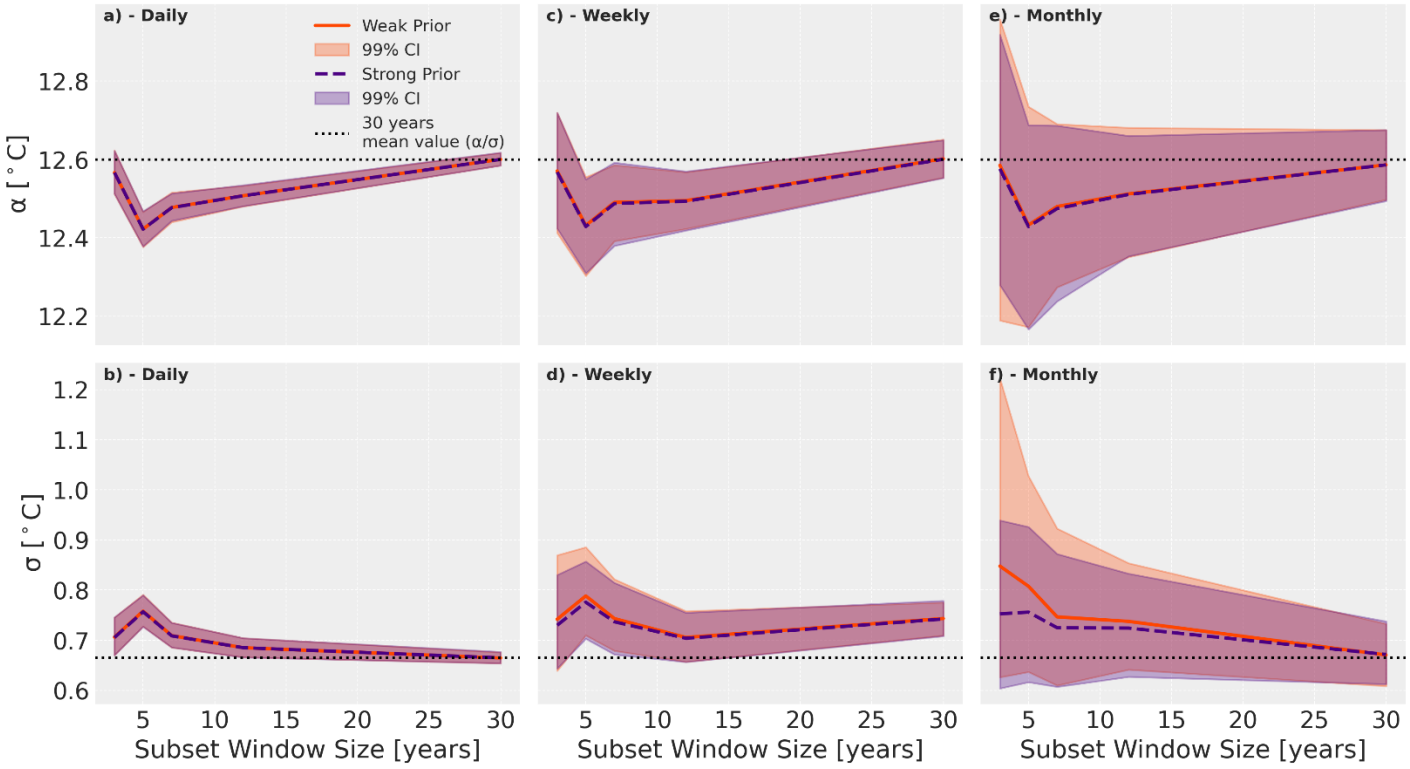


Figure S3.5. Same as Figure 6 but for site UK.

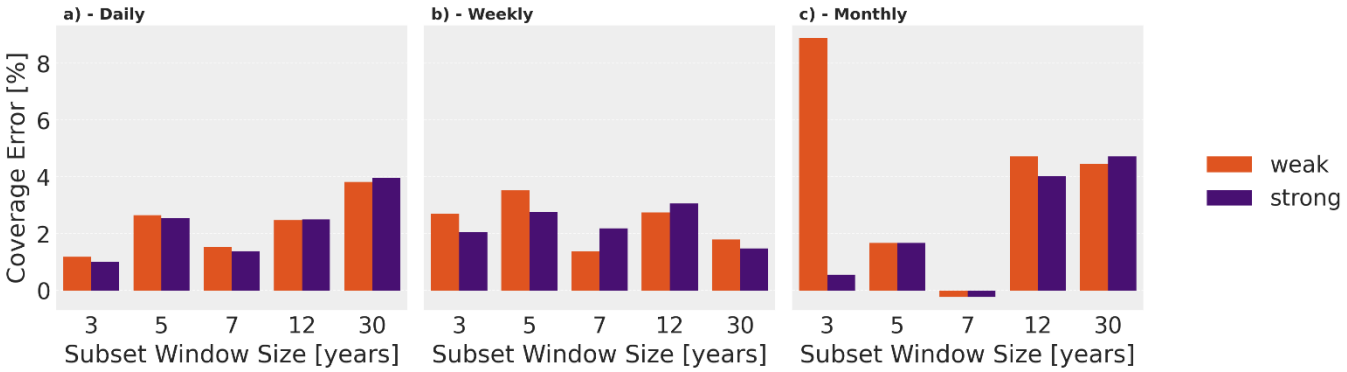


Figure S3.6. Same as Figure 7 but for site UK.

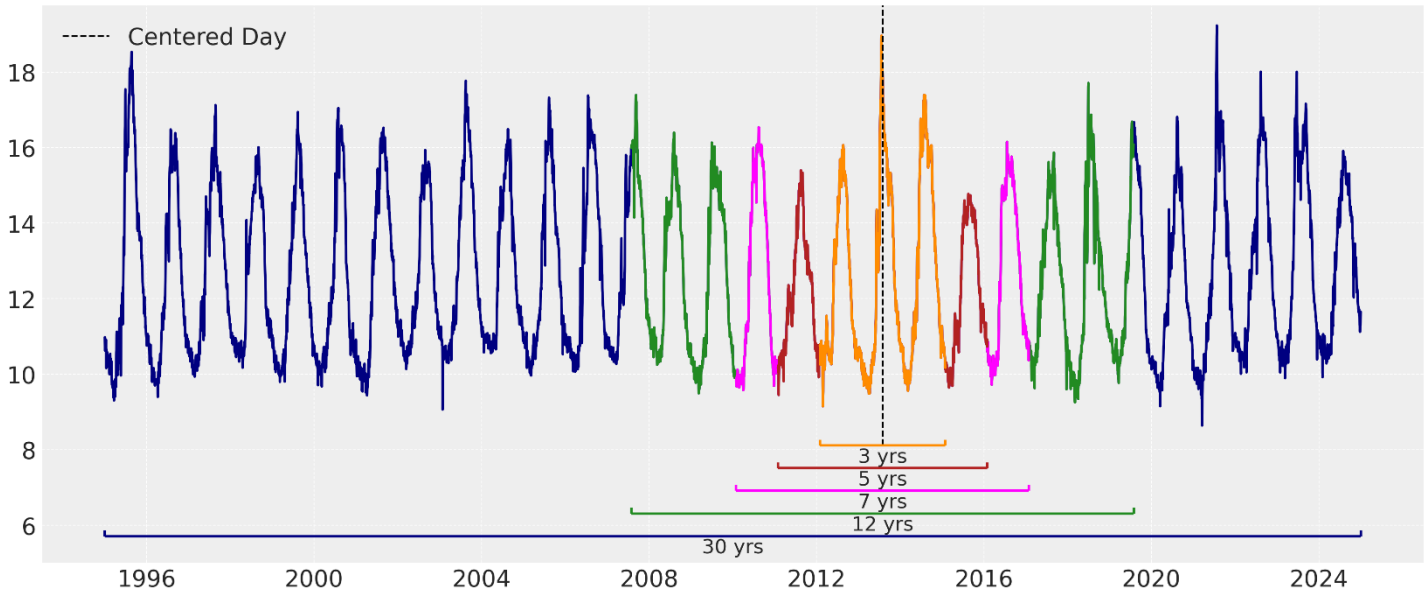


Figure S3.7. Same as Figure 8 but for site UK.

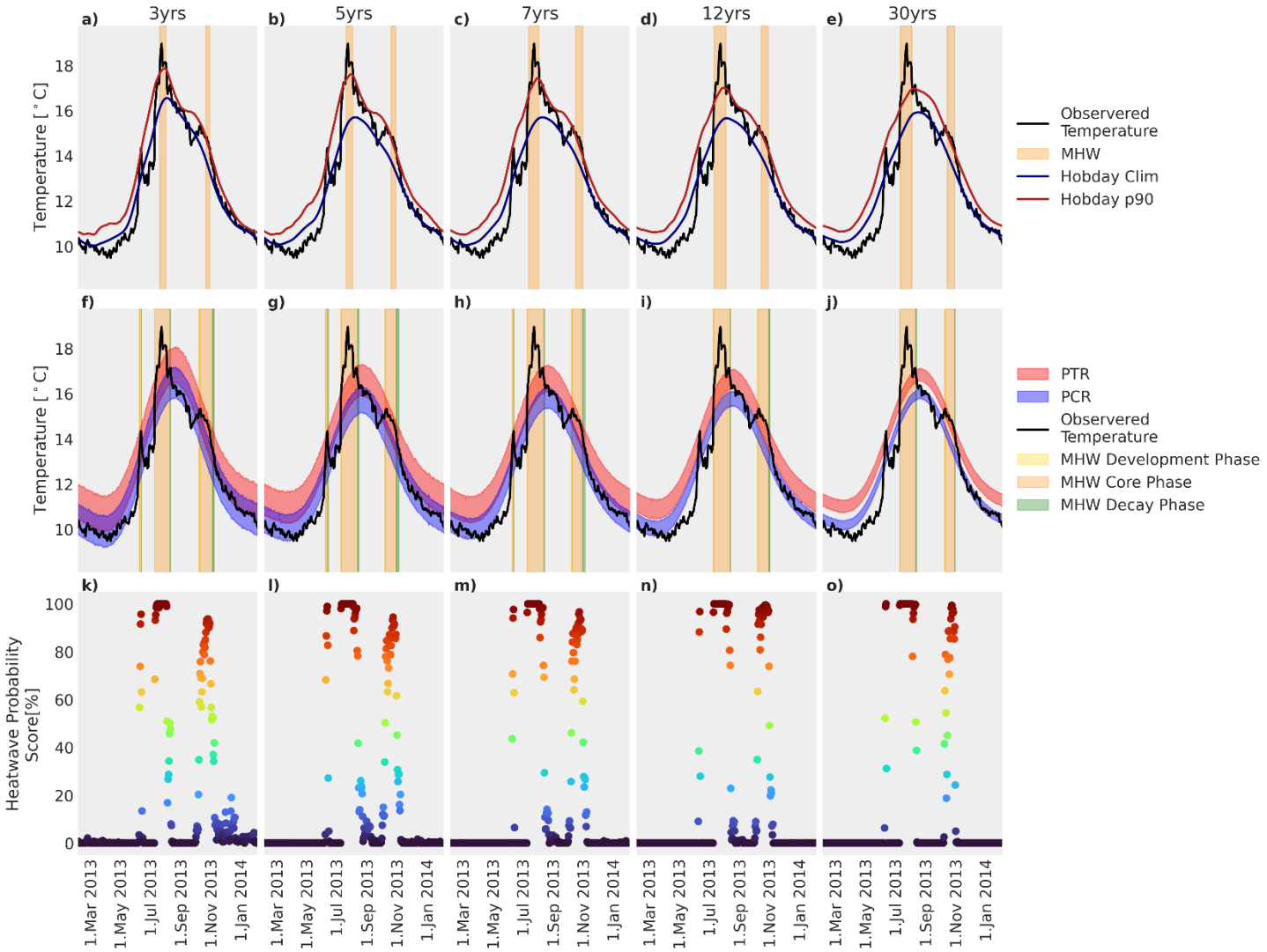


Figure S3.8. Same as Figure 9 but for site UK.

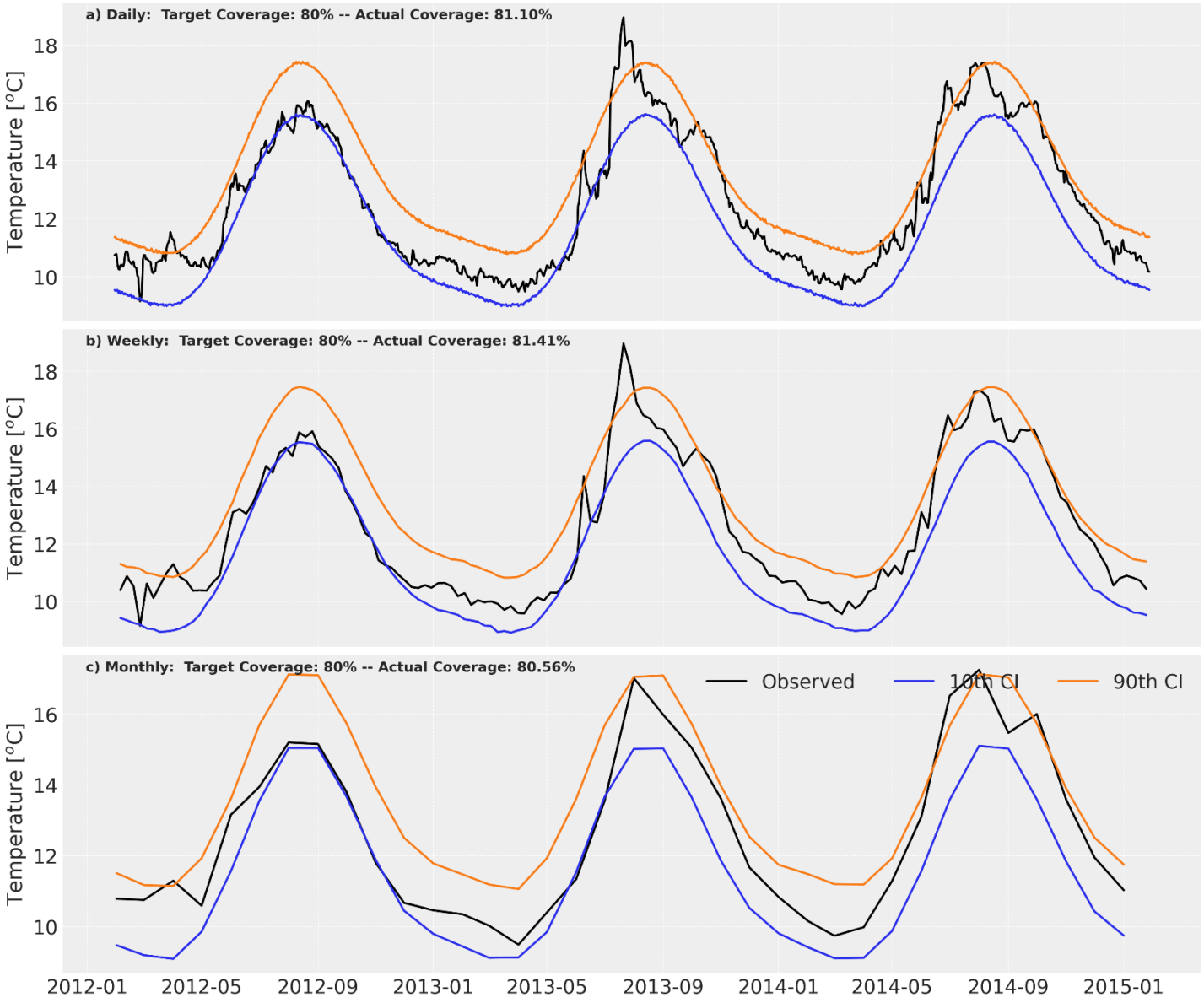


Figure S3.9 Same as Figure S0.3 but for site UK.

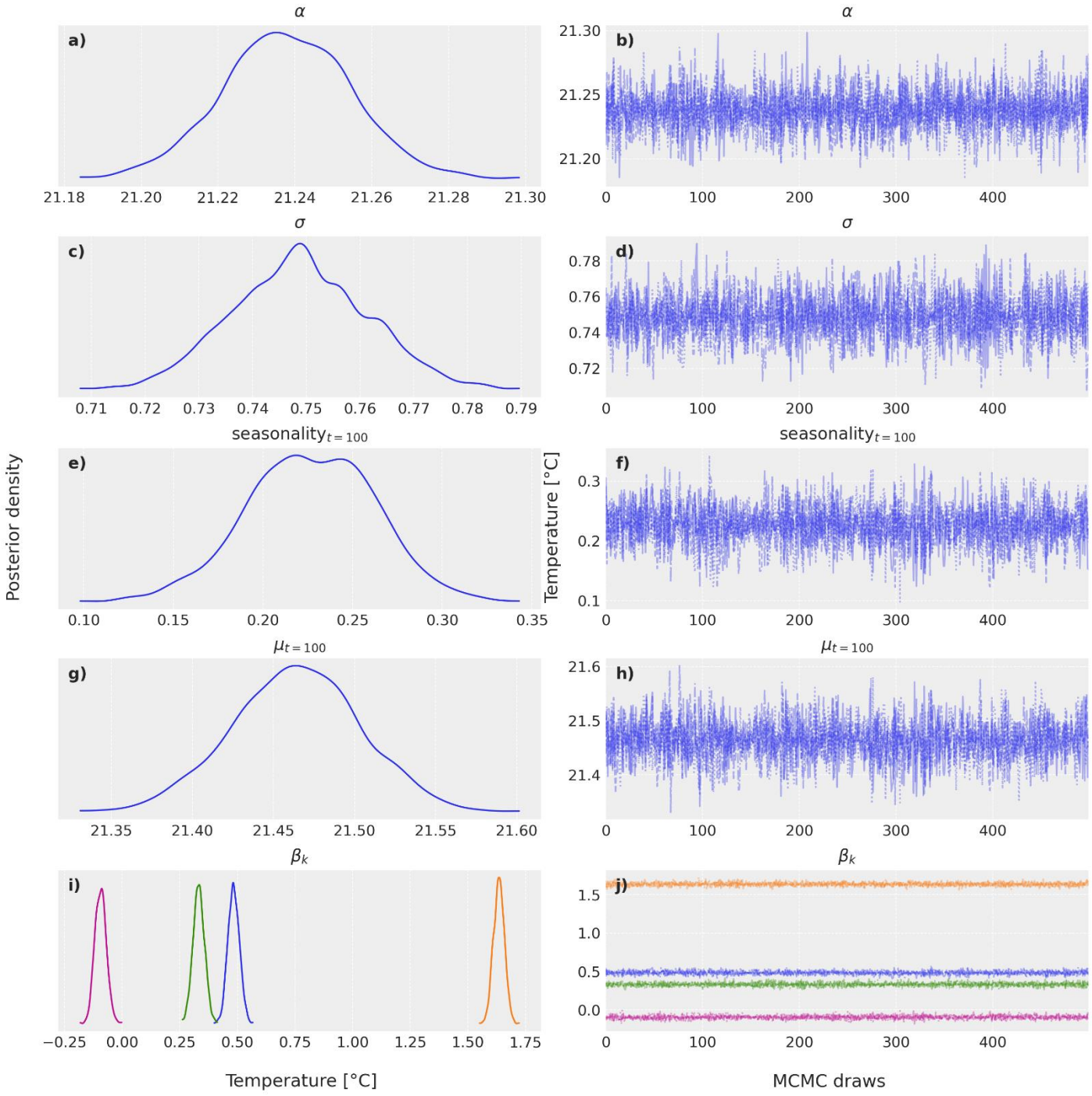


Figure S4.1. Same as Figure 4 but for site SouthWA 50m**.**

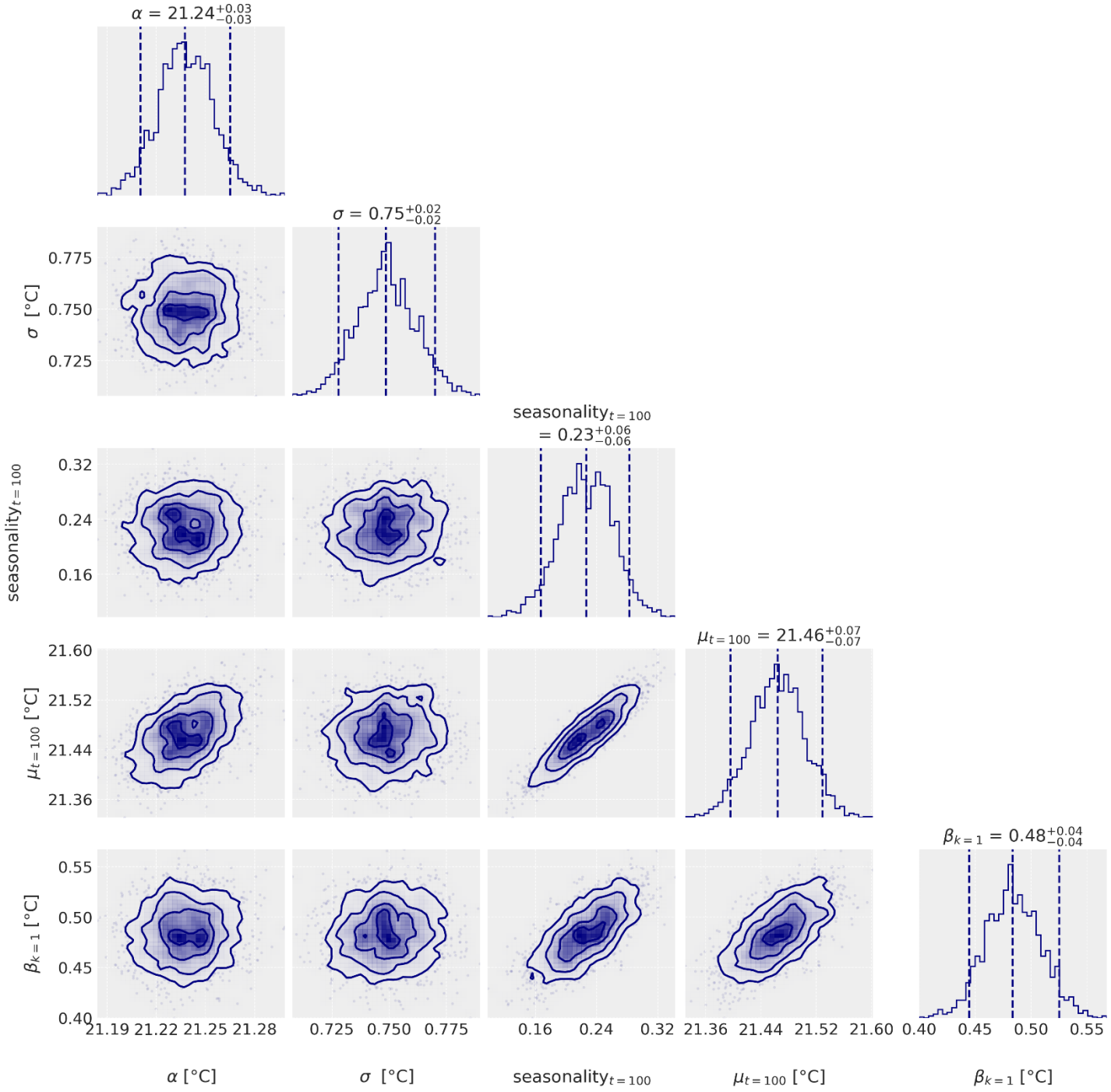


Figure S4.2. Same as Figure S0.1 but for site SouthWA 50m**.**

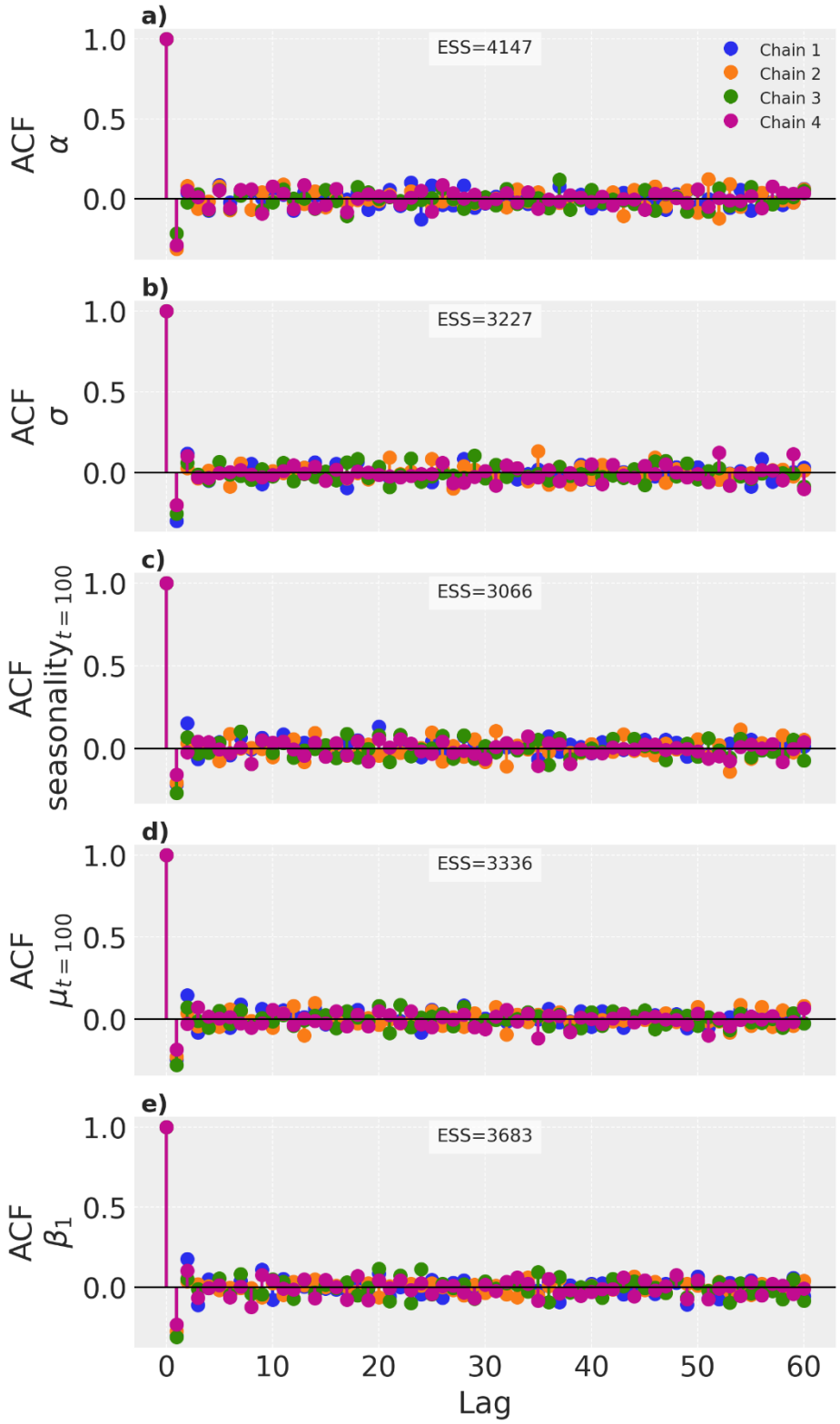


Figure S4.3. Same as Figure S0.2 but for site SouthWA 50m**.**

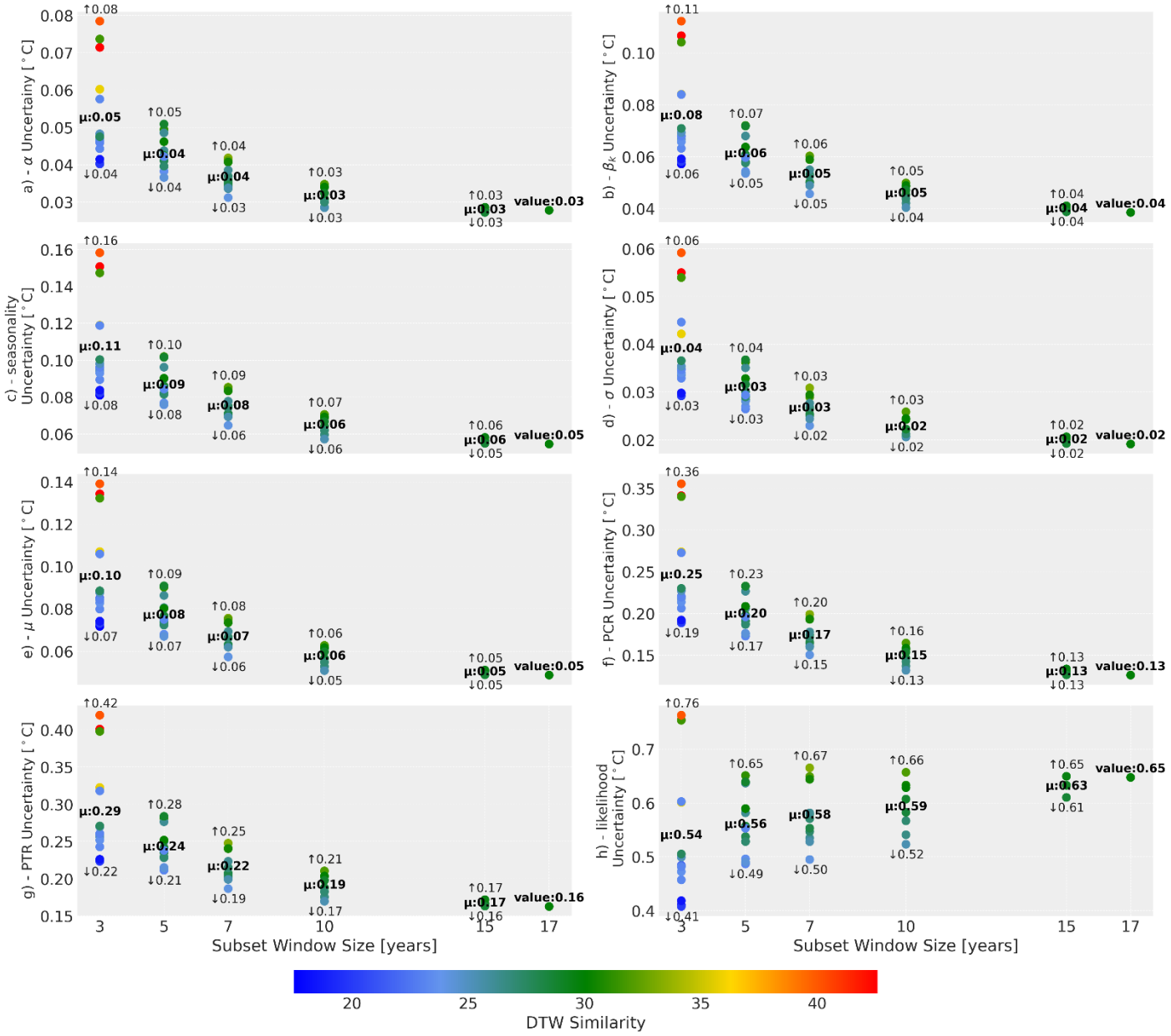


Figure S4.4. Same as Figure 5 but for site SouthWA 50m**.**

Table S4.4. same as Table 2 but for 17 years of data for sihte SouthW**A** 50m.

Subset Length [years]	3	5	7	10	15	17
Number of Subsets	15	13	11	8	3	1

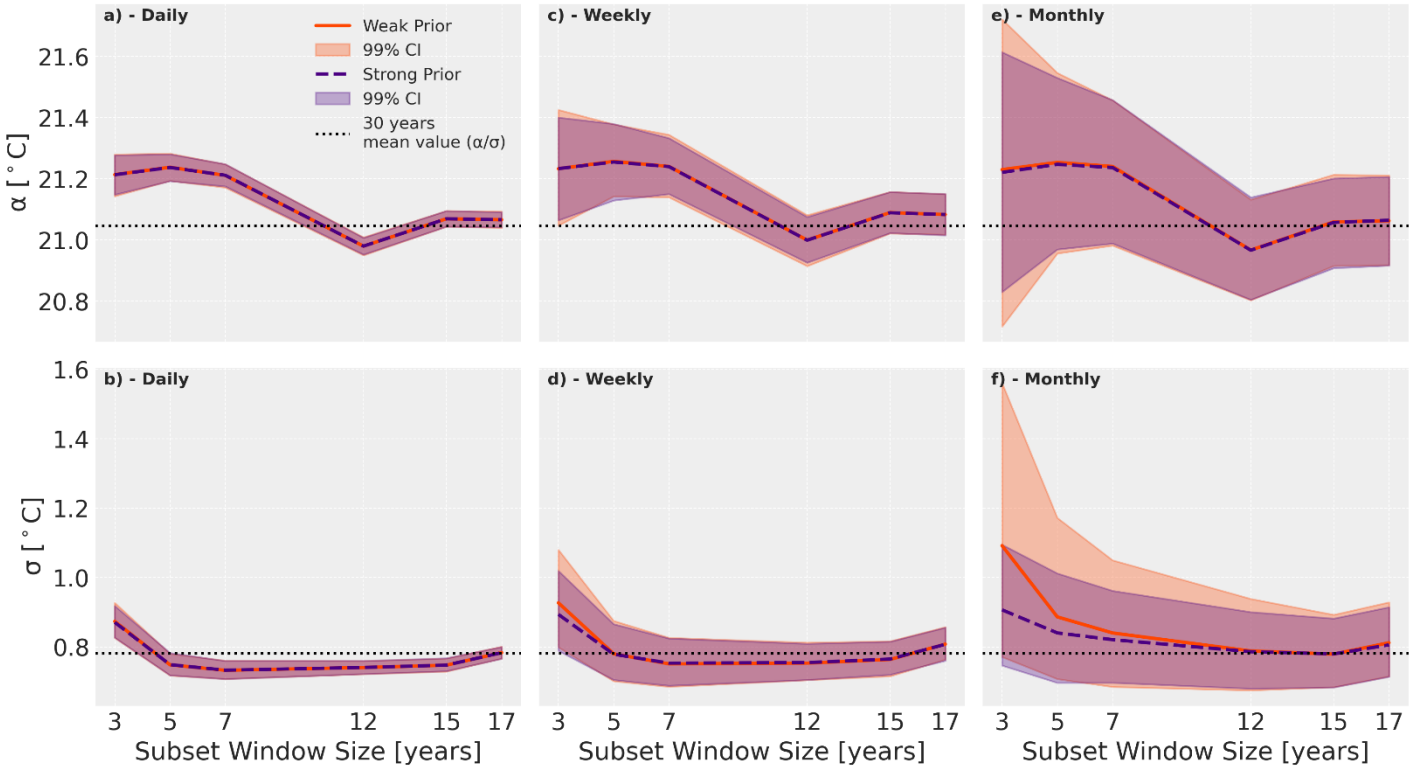


Figure S4.5. Same as Figure 6 but for site SouthWA 50m**.**

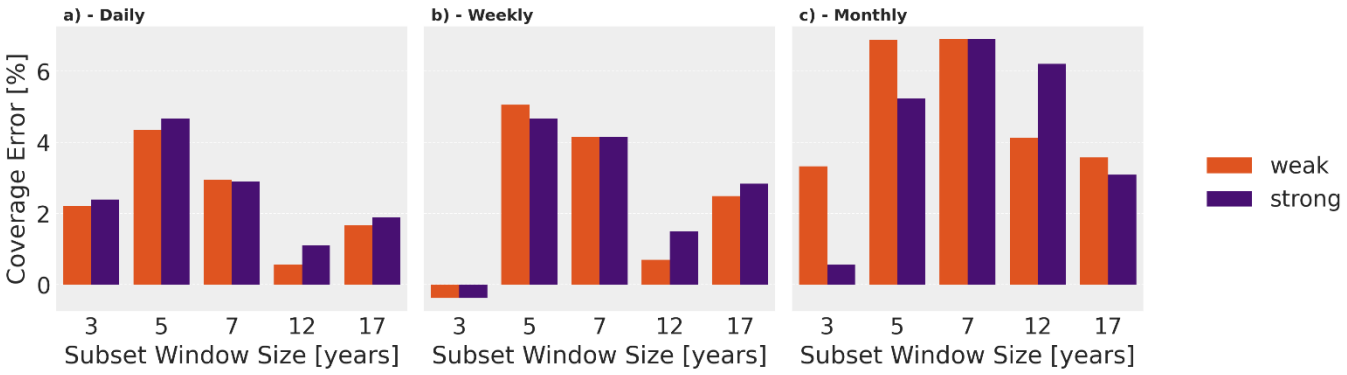


Figure S4.6. Same as Figure 7 but for site SouthWA 50m.

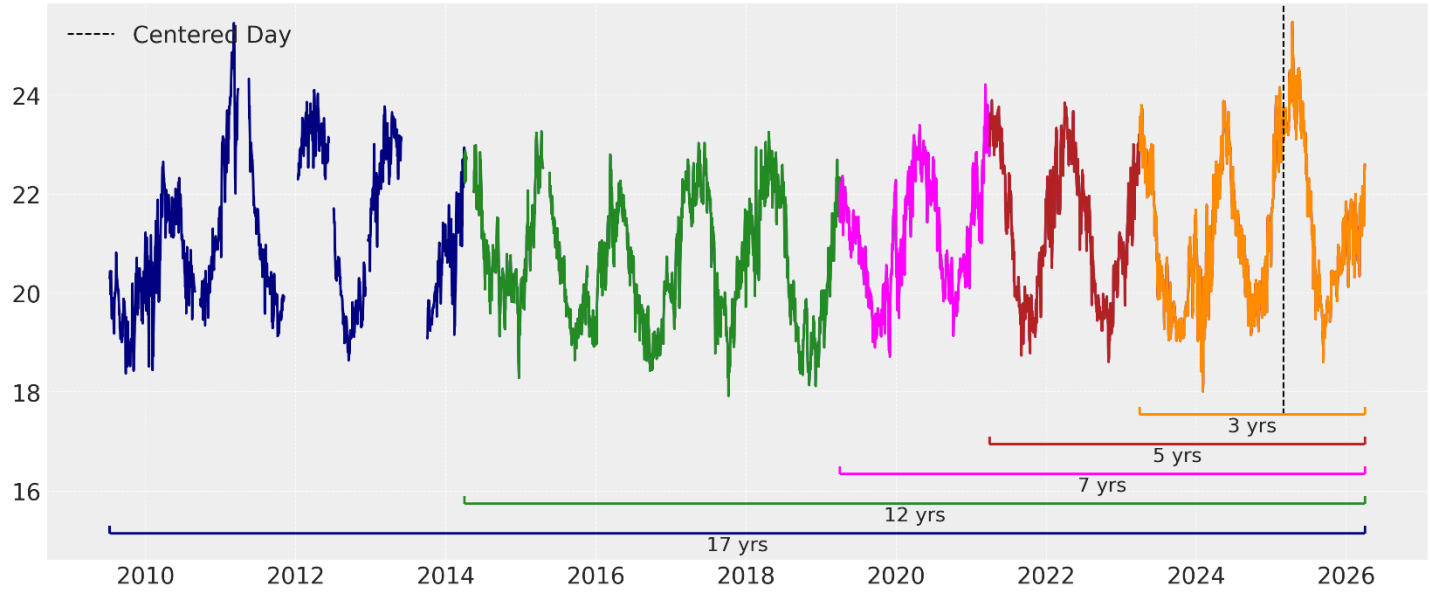


Figure S4.7. Same as Figure 8 but for site SouthWA 50 m.

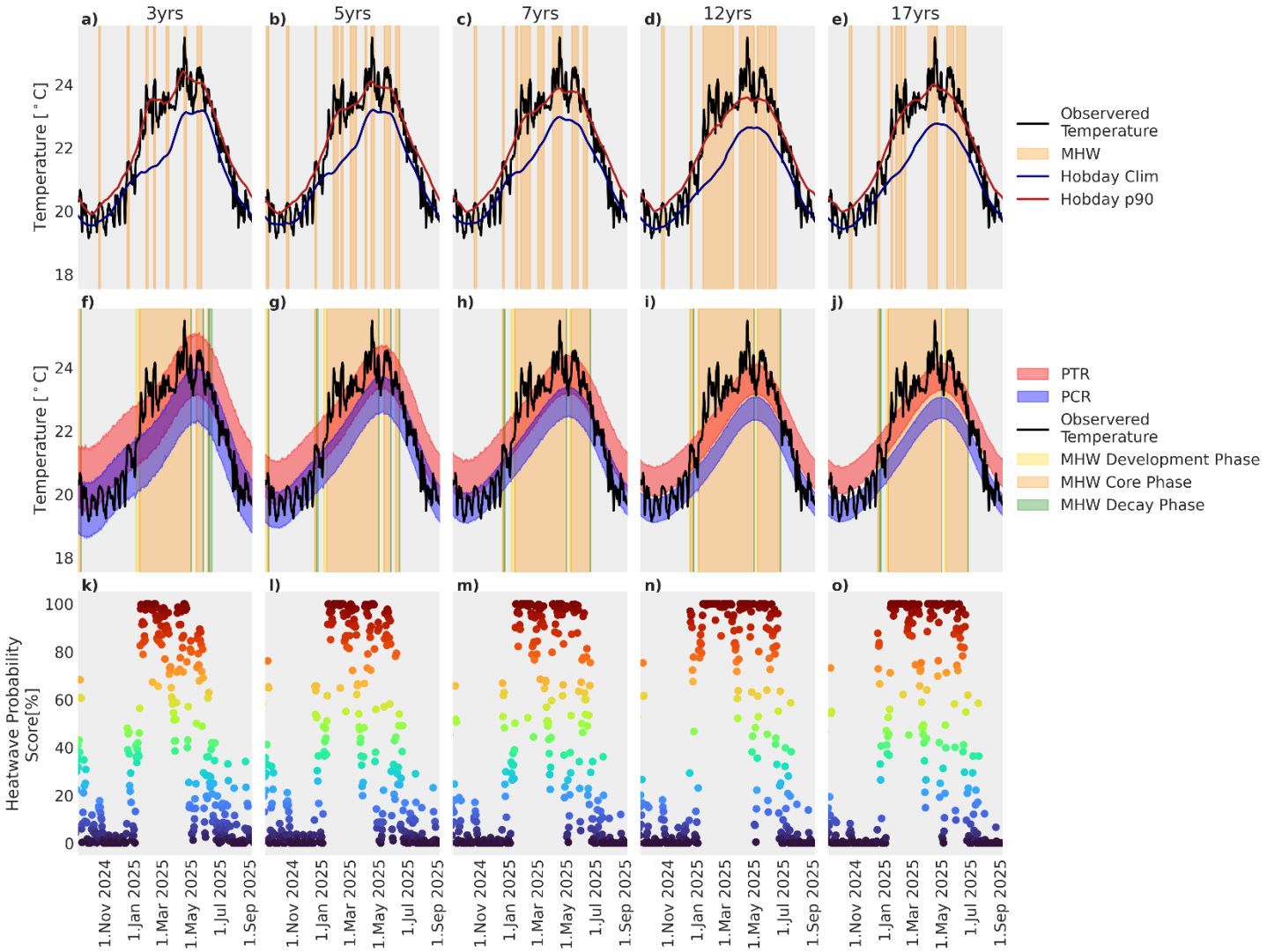


Figure S4.8. Same as Figure 9 but for site SouthWA 50 m.

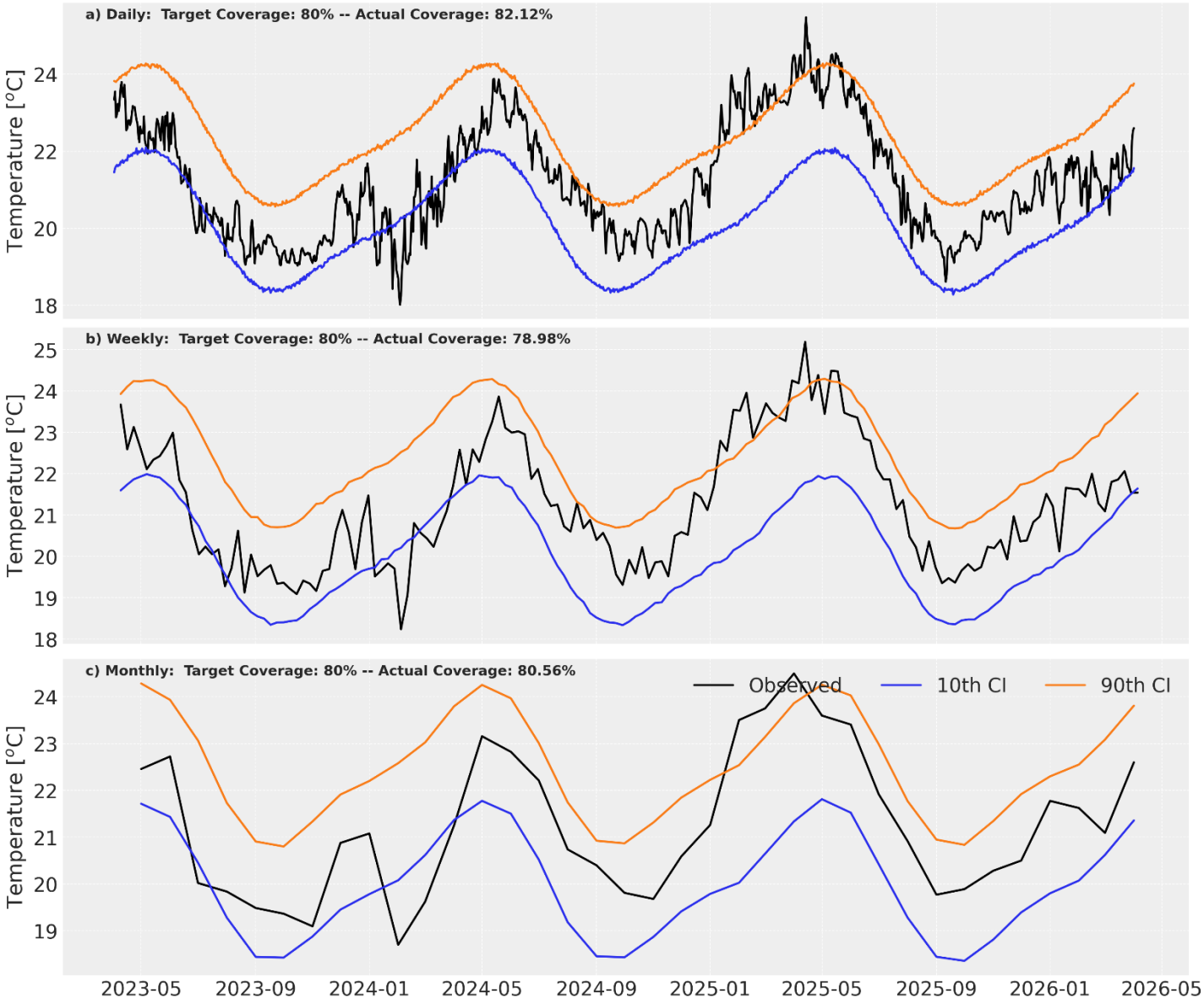


Figure S4.9 Same as Figure S0.3 but for site SouthWA 50 m.